

# Persistence Mechanics of the Quantum Vacuum III

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When a persistent system's internal complexity exceeds the capacity of a single control loop, it must structurally partition or collapse. We present three universal rules governing this partitioning: Capacity Partitioning (distributing finite bandwidth across orthogonal demands), Reversible Encoding (conserving non-integer mismatches to avoid Landauer erasure), and Impedance Matching (minimizing reflection loss between nested layers). These rules bridge the gap between architectural requirements (what persistent systems must be) and operational mechanics (how they continue to persist). Applying these rules to the  $E_8$  substrate derived in companion works, we show that the constants of the Standard Model and General Relativity emerge as the exact, parameter-free capacity allocations. Gauge couplings arise as bandwidth fractions under Capacity Partitioning (the strong coupling, QCD beta function coefficients, weak mixing angle, and Jarlskog invariant). Electroweak mass scales arise as impedance-matching thresholds under Impedance Matching at the Resonant Interface (the Higgs vacuum expectation value, Fermi constant,  $W$ -boson mass, and Higgs mass). Gravitational scales arise as the conserved geometric attenuation of Reversible Encoding across the Geometric Depth (the Planck mass and vacuum energy density), simultaneously resolving the Gravitational Hierarchy and the Vacuum Catastrophe. A zero-degree-of-freedom global geometric fit spanning 64 orders of magnitude ( $\alpha^{-1}$ ,  $G_F$ ,  $M_W$ ,  $\alpha_s$ ,  $M_H$ ,  $\sin^2 \theta_W$ ,  $J$ ,  $M_P$ ,  $R_K$ ) yields a reduced  $\chi^2 \approx 1.1687$ .

## I. INTRODUCTION

In *Informational Energetics: A Universal Architecture of Persistence* [1], we introduced Informational Energetics (IE) and established that any persistent system must resist entropic decay by satisfying six architectural pillars derived from control theory, information theory, and thermodynamics: Capacity (the vessel), Map (the model), Protocol (the interface), Governor (the constraint), Toll (the temporal cost), and Margin (the buffer). In *Informational Energetics: Entropic Action* [2], we formalized the Entropic Action ( $S_\Phi$ ) which defines the total thermodynamic cost for a persistent system to maintain structural coherence.

While previous work established what a persistent system must *be* and how it must *behave*, neither addressed how it can continue to persist when its regulatory complexity exceeds the capacity of a single control loop. Subpartitioning can be found in every domain: a biological cell that must simultaneously maintain membrane integrity, metabolize, and replicate partitions into organelles; a corporation partitions into departments. In section II, we define three fundamental rules of **Nested Persistence** governing this partitioning: *Capacity Partitioning*, *Reversible Encoding*, and *Impedance Matching*. Together, they define how a system partitions into specialized subsystems to ensure continued persistence.

## A. The Test: The Vacuum

The vacuum is the ideal test case. Macroscopic systems frequently utilize external energy for partitioning, but a closed system with no external environment to offset its dissipation is forced to solve the partitioning problem with zero error. It therefore provides the most stringent possible test of whether these rules are universal.

To test these rules, we apply them to the  $E_8$  substrate derived in [1]. The functional operator forms were established in [2], but their numerical coefficients remained undetermined. When the rules of Nested Persistence are applied, they emerge as coupled partition coefficients of a single finite capacity budget. Continuing the numbering established in the companion papers (System 0: architectural requirements, System 1: the substrate, System 2: geometric impedance, and System 3: Entropic Lagrangian dynamics):

- **System 4: The Gauge Partition** (Rule 1: Capacity Partitioning). The  $\nu = 16$  chiral capacity is subdivided into non-overlapping geometric allocations, yielding the gauge couplings: Strong Coupling ( $\alpha_s$ ), Weak Mixing Angle ( $\sin^2 \theta_W$ ), and Jarlskog Invariant ( $J$ ). Because the temporal and color sectors are geometrically isolated by this partition, we show that the Strong CP problem is structurally prohibited without requiring an axion.
- **System 5: The Resonant Interface** (Rule 3: Impedance Matching). A nested subsystem stabilizing local topological boundaries against vacuum resonance, identified as the Higgs mechanism. Generates the electroweak mass scale: Higgs VEV ( $v$ ), Fermi constant ( $G_F$ ), and Electron Yukawa ( $y_e$ ).
- **System 6: The Geometric Depth** (Rules 1 & 2:

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Capacity Partitioning and Reversible Encoding). A nested subsystem stabilizing the macroscopic lattice volume, identified as Gravity. Signal attenuation derives the Planck scale ( $M_P$ ) and Cosmological Constant ( $\Lambda$ ), which we show accounts for the observed Gravitational Hierarchy and Vacuum Catastrophe.

A zero-degree-of-freedom global geometric fit spanning 64 orders of magnitude ( $\alpha^{-1}$ ,  $G_F$ ,  $M_W$ ,  $\alpha_s$ ,  $M_H$ ,  $\sin^2 \theta_W$ ,  $J$ ,  $M_P$ ,  $R_K$ ) yields a reduced  $\chi^2 \approx 1.1687$ . This convergence demonstrates that the fundamental constants are the structurally necessary partition coefficients of a finite information substrate.

## II. ARCHITECTURE OF NESTED PERSISTENCE

Every finite information-processing system such as a cell, corporation, or physical substrate hits a hard capacity ceiling. When entropic demands multiply (internal maintenance, external coordination, or both), the regulatory burden on any single loop grows. Once that burden exceeds capacity, the system's internal model loses track of reality and collapses. Only systems that partition persist. They do so by splitting into specialized subsystems, each managing a tractable fraction of the total threat space. This partitioning is governed by three operations: Capacity Partitioning splits the budget across orthogonal demands, Reversible Encoding conserves the indivisible remainder without erasure, and Impedance Matching binds the nested layers without reflection loss.

**Example:** Consider a firm with a finite operational budget of \$10M that must simultaneously manufacture, innovate, and comply with regulations. *Capacity Partitioning* requires splitting the \$10M across these three departments; the sum of their allocations plus accounting overhead must exhaust the total. When the budget does not divide evenly, \$3M per department leaves a \$1.0M remainder, *Reversible Encoding* requires carrying this indivisible remainder as a strategic reserve rather than discarding it (which would incur an audit penalty). Simultaneously, *Impedance Matching* resolves the friction when an executive team generates initiatives at a higher frequency than R&D can absorb; a middle manager acts as a transformer, scaling and scheduling the executive signal to match the departmental aperture and prevent reflection loss (missed deadlines, dropped projects).

### A. Capacity Partitioning

**Foundational Principle.** By Ashby's Law of Requisite Variety [3], a control system must possess at least as many states as the perturbations it seeks to regulate. However, the maximum rate at which a system can transition between states is strictly bounded by its physical capacity [4].

**Activation Threshold.** When the number of concurrent regulatory demands requires a computation rate exceeding what a single loop can execute within its fundamental update cycle, the system must split into parallel, nested subsystems.

**Operational Mandate.** The system must partition its resources such that the sum of the active subsystem capacities exhausts the total budget.

**Structural Cost.** Creating a boundary between subsystems inherently consumes capacity (synchronization overhead, data framing, interface maintenance). This **irreducible structural overhead** is the cost of the boundary itself.

### B. Reversible Encoding

**Foundational Principle.** Information erasure incurs an irreducible thermodynamic tax (Landauer's Principle [5]), which would compound and destabilize the system if the remainder of any partition were discarded.

**Activation Threshold.** When a system's workload cannot divide evenly across its subsystems, this structural mismatch leaves a remainder that cannot be discarded.

**Operational Mandate.** The system must perform **Reversible Encoding** [6]: it must strictly conserve the mismatch by storing it as structural metadata.

**Structural Cost.** The metadata itself consumes capacity. A ledger that carries a rounding error must allocate bits to that remainder; a scheduling queue that carries a temporal offset must allocate slots to that offset. This storage overhead is the price of avoiding erasure.

### C. Impedance Matching

**Foundational Principle.** When a system partitions its capacity, it creates nested subsystems that operate on different geometric or functional scales. Information crossing the boundary must be translated. To minimize total Entropic Action ( $S_\Phi$ ) [2] across the hierarchy, these translations must minimize reflection loss (Shannon [7]).

**Activation Threshold.** When nested subsystems operate at different scales, any signal crossing the boundary risks reflection.

**Operational Mandate.** The internal processing resistance of a subsystem must closely match the geometric bandwidth allocated to it by the partition.

**Structural Cost.** If there is a mismatch, the signal translation fails, information reflects at the boundary, and the resulting excess entropy causes the system to decay. To prevent this, the system must engineer the match, consuming capacity to align internal resistance with external aperture.

### D. Recursive Universality

These three rules apply recursively: each partitioned subsystem is itself a persistent entity, instantiating the full six-pillar architecture on its fraction of the parent's capacity. If its allocated capacity is insufficient to regulate its own internal complexity it must partition again. The full architecture of a complex persistent system is therefore a **finite nesting hierarchy** (a nearly decomposable system, *cf.* Simon [8]), with each level's partition coefficients strictly determined by the parent level's bandwidth allocation. No level introduces free parameters; each inherits its budget from above and exhausts it completely.

This recursion is universal: a multicellular organism partitions metabolic bandwidth across organ systems; a corporation partitions operational capacity across departments. In every domain, the partition coefficients must satisfy the global solvency constraint of nested persistence. This paper applies these principles strictly to the quantum vacuum; their application to macroscopic domains is the subject of future work.

### III. THE PHYSICS VALIDATION: THE $E_8$ SUBSTRATE AND THE GEOMETRIC CASCADE

As derived in *Informational Energetics: A Universal Architecture of Persistence* [1], the vacuum is the unique resource-constrained computational substrate satisfying the six architectural pillars of persistence: an  $E_8$  lattice projected to  $D = 4$  spacetime, with each localized node possessing strictly finite channel capacity ( $\nu = 16$ ). This substrate contains four key invariants: the temporal resonance  $\Delta = 43$ , the chiral capacity  $\nu = 16$ , the interaction rank  $\sigma = 5$ , and the topological boundary  $\chi = 2$ , all derived from the  $E_8 \rightarrow D_4 \oplus D_4$  projection.

While we derived the functional operator forms in *Informational Energetics: Entropic Action* [2], the numerical coefficients remained undetermined. Standard QFT treats these constants as independent empirical inputs, each determined by measurement. In the IE framework, they emerge as coupled partition coefficients of a single finite capacity budget. Unlike  $\alpha^{-1}$ , which represents the unpartitioned baseline impedance derived directly from the substrate, the remaining parameters are not fundamental. They emerge strictly as the partition coefficients of the vacuum's finite channel capacity ( $\nu = 16$ ): the unique allocation required for nested subsystems to satisfy global solvency.

If the spatial geometry, temporal updates, and internal force generators all attempt to saturate the node simultaneously, local complexity exceeds the 16 available channels and the causal structure becomes unresolvable. Because the vacuum persists, its structure must satisfy the three rules of Nested Persistence (section II). The  $\nu = 16$  budget is therefore partitioned across orthogonal interaction sectors:

**Rule 1 (Capacity Partitioning)** allocates the  $\nu = 16$  chiral capacity across orthogonal geometric sectors. This yields **System 4: The Gauge Partition**: the saturation ratio (Strong Coupling  $\alpha_s$ ), the spacetime split (Weak Mixing Angle  $\sin^2 \theta_W$ ), and the irreducible geometric frustration of projecting 5-fold internal symmetry onto 4-fold spacetime (Jarlskog Invariant  $J$ ). Because the temporal and color sectors are geometrically isolated by this partition, the Strong CP problem is structurally prohibited without requiring an axion.

**Rule 3 (Impedance Matching)** stabilizes the boundary between the high-frequency vacuum resonance and local topological defects. This yields **System 5: The Resonant Interface**, generating the electroweak mass scale: the Higgs VEV ( $v$ ), Fermi constant ( $G_F$ ), and the minimum resolvable mass (Electron Yukawa  $y_e$ ), the floor below which vacuum fluctuations erase distinguishable states.

**Rules 1 & 2 (Capacity Partitioning and Reversible Encoding)** govern the residual bandwidth, now attenuated across the geometric bulk. This yields **System 6: The Geometric Depth**, identified as Gravity. The attenuation depth derives the Planck scale ( $M_P$ ); the irreducible bulk resolution floor derives the Vacuum Energy Density ( $\rho_{vac}$ ), resolving both the Gravitational Hierarchy and the Vacuum Catastrophe.

#### A. Note on Absolute Scale

A discrete lattice inherently produces pure dimensionless ratios:  $y_e = m_e/v$ ,  $v/m_e \approx 4.8 \times 10^5$ ,  $M_P/m_e \approx 2.4 \times 10^{22}$ . These ratios are the framework's genuine zero-parameter predictions. To express them as absolute values in laboratory units (MeV, GeV), one scale must be fixed — exactly as expressing the speed of light in m/s requires the definition of a meter. This anchoring is intrinsic to any scale-invariant framework and introduces no free parameter.

We adopt  $m_e \approx 0.511$  MeV as the **natural unit anchor**. This choice is not arbitrary: the electron mass is the **Landauer Limit** of the substrate, derived below in section V E as the Margin output of System 5. It is the minimum resolvable topological defect — any lighter charged state would dissolve into vacuum thermal noise. The electron is therefore the *structural floor* that the partition rules necessarily produce; it is the natural unit in the same sense that Planck units are natural for quantum gravity.

All higher mass scales ( $v$ ,  $M_W$ ,  $M_H$ ,  $M_P$ ) follow as strict geometric multiples of this floor. The derivations do not use  $m_e = 0.511$  MeV as a numerical input; they derive the dimensionless ratios, and the single anchor converts them to laboratory units. This cascade spans 64 orders of magnitude from the gravitational coupling ( $\alpha_G \sim 10^{-45}$ ) to the Planck mass ( $M_P \sim 10^{19}$  GeV) with **zero tunable parameters**.

#### IV. SYSTEM 4: THE GAUGE PARTITION

Under Rule 1 (**Capacity Partitioning**), a discrete substrate with finite node capacity ( $\nu = 16$ ) faces an immediate structural bottleneck: if spatial translations, temporal updates, and internal force mediation attempt to utilize the channels simultaneously, the local state space overflows. This results in Causal Aliasing: when a node attempts to process spatial translations ( $D = 4$ ), temporal updates ( $\Delta = 43$ ), and internal force mediation ( $\sigma = 5$ ) through the same finite channel set, the total state transitions per cycle exceed the  $\nu = 16$  capacity. The local causal structure loses track of which information belongs to which process, and the node decoheres.

To persist, the capacity must be split into non-overlapping fractional bandwidths for each orthogonal demand, creating specialized subsystems to manage them. This routing problem defines **System 4: The Gauge Partition**. When evaluated, these structural limits identically match the core empirical parameters of the Standard Model gauge sector:

1. **Capacity: The Chiral Bandwidth** ( $\nu = 16$ ). The  $E_8$  substrate possesses 16 chiral degrees of freedom. This acts as the absolute processing budget that must be distributed across all tasks without overflow. Because each subsystem must share a common coordinate grid to remain causally coherent, any signal traversing the bulk manifold incurs the Coordinate Overhead ( $\eta = 171/172$ ). When all channels are active, the resulting impedance load is  $\alpha_s$ .
2. **Map: The Gauge Topology** ( $\sigma = 5, \chi = 2$ ). To embed a distinct topological defect into the substrate, the system must map its internal interaction generators (rank  $\sigma = 5$ ) and localized boundaries ( $\chi = 2$ ) onto the available capacity. These invariants define the number of distinct interaction apertures requiring bandwidth.
3. **Protocol: The Spacetime Partition** ( $\sin^2 \theta_W$ ). The bandwidth fraction allocated to temporal updates versus spatial structure that prevents serialization deadlock.
4. **Governor: Field Rigidity** ( $\beta_0$ ). The geometric resistance to gauge field deformation that prevents partition ratios from warping under perturbation.
5. **Toll: Projection Frustration** ( $J$ ). The irreducible geometric mismatch cost of mapping 5-fold symmetry onto 4-fold spacetime.
6. **Margin: Minimum Resolvable Impedance** ( $y_e$ ). The structural noise floor below which vacuum fluctuations erase distinguishable states. (The

Electron Yukawa coupling explicit derivation is deferred to section V, where it emerges as the minimum resolvable impedance of the resonant interface.)

##### A. Capacity & The Coordinate Overhead ( $\eta$ )

Under Rule 1 (Capacity Partitioning), splitting the  $\nu = 16$  budget into orthogonal subsystems creates multiple independent signal paths. After the partition, signals must traverse a shared coordinate grid to reach their designated subsystem. This coordination requires a frame delimiter that separates control information from payload. On a discrete lattice, this delimiter is the coordinate origin: the site occupied by the reference frame that cannot simultaneously carry physical state information.

This overhead applies exclusively to *active signal transmission* over the bulk manifold volume (such as the channel saturation  $\alpha_s$ ). Quantities defined by boundary topology (1-forms, Wilson Loops,  $\chi$ ) do not reference the internal coordinate grid and are strictly unattenuated by  $\eta$ .

The substrate's maximum address space per cycle is the Manifold Resolution  $R_M = D \times \Delta = 4 \times 43 = 172$ . The origin consumes exactly one slot from this finite budget, leaving  $N_{\text{addressable}} = R_M - 1 = 171$  addressable payload sites. No other allocation is permitted:

- $N_{\text{addressable}} = 172$  (no overhead): Requires a continuous manifold with infinite information density, violating Finiteness.
- $N_{\text{addressable}} = R_M - k$  for integer  $k > 1$ : Requires  $k$  independent reference frames. The Closure Constraint forbids external references; only the self-referential origin is permitted.
- $N_{\text{addressable}} = 0$  (total overhead): No addressable capacity remains; the substrate cannot process information.
- $\eta$  dependent on  $\Delta$ : The temporal resonance is a periodic cycle with no endpoints; it requires no origin fixing. Applying  $\eta$  to  $\Delta$  would double-count the structural cost.

The Coordinate Overhead is therefore the strict geometric ratio of addressable payload capacity to total hardware limit:

$$\eta \equiv \frac{N_{\text{addressable}}}{N_{\text{total}}} = \frac{R_M - 1}{R_M} = \frac{171}{172} \approx 0.9941860465116279 \quad (1)$$

The numerator captures the state's information load (Capacity and Protocol); the denominator is the full Geometric Impedance of the substrate, incorporating all six pillars.

## B. Capacity Saturation (The Strong Coupling $\alpha_s$ )

The Fine-Structure Constant ( $\alpha^{-1}$ ) is the baseline impedance of the vacuum: the resistance to a single-channel transmission derived directly from the  $E_8$  substrate geometry [1]. The Capacity Partition, however, requires a subsystem that saturates every available chiral channel to enforce color confinement. When such a fully confined state traverses the bulk manifold, it imposes a load on the network proportional to all  $\nu = 16$  channels simultaneously. The **Strong Coupling** ( $\alpha_s$ ) emerges as the ratio of this saturated load to the baseline geometric impedance:

1. **Chiral Saturation** ( $\nu \cdot \eta$ ): The defect must actively drive all  $\nu = 16$  chiral channels to maintain internal coherence. Because this confined state represents active payload propagating through the discrete bulk manifold, it incurs the Coordinate Overhead ( $\eta$ ).
2. **Isotropic Flux** ( $1/D$ ): The defect must anchor to the 4-dimensional spatial grid. By the Homogeneity constraint, a single unit of flux must distribute equally across the  $D = 4$  orthogonal spacetime directions, yielding a normalization factor of strictly  $1/D = 1/4$ .

The maximum operational efficiency of this confined state is the ratio of this Total Active State Load to the Geometric Impedance ( $\alpha^{-1}$ ):

$$\alpha_s = \frac{\text{State Load}}{\text{Impedance}} = \frac{\nu \cdot \eta + 1/D}{\alpha^{-1}} \quad (2)$$

### 1. Verification: The Strong Coupling at the Z-Pole

The strong coupling ( $\alpha_s$ ) can change value or “runs” depending on the energy scale of the interaction. To test our geometric derivation, we must evaluate it at the appropriate physical limit. In experimental physics, the saturation threshold is observed as the Electroweak Resonant Scale (the Z-Pole,  $M_Z$ ). This is the specific physical energy density where the vacuum’s internal capacity is fully “illuminated” before macroscopic spatial dilution (running) sets in.

Substituting the geometric invariants yields the exact physical magnitude of the Strong Coupling at this saturation limit:

$$\alpha_s(M_Z) = \frac{16 \cdot \eta + 1/4}{\alpha^{-1}} \approx \mathbf{0.117903156} \quad (3)$$

**Prediction:** 0.117903156

**Experiment:**  $0.1179 \pm 0.0009$  [9]

**Accuracy:**  $0.00\sigma$

## C. Rigidity: The Beta Function ( $\beta_0$ )

The **Governor** pillar of the Capacity Partition requires structural rigidity: the partition ratios must resist deformation under localized perturbations. The QCD beta function coefficients are the exact geometric encoding of this stiffness.

*a. Geometric Stiffness: The Anti-Screening Term*  
The coefficient “11” is the trace of the rigidity operator on the local Hilbert space. It is forced by the tensor product structure of orthogonal perturbations:

The Spacetime Anchor ( $D \times \chi$ ) and Symmetry Pressure ( $\sigma - \chi$ ) act on geometrically independent subspaces (spacetime vs. internal). Their contributions combine via tensor product, not direct sum, because the perturbations are orthogonal. The trace of a tensor product is the product of traces; here each factor has trace equal to its rank, yielding  $8 + 3 = 11$ . Alternative combinations violate this orthogonality:  $D \times \sigma = 20$  double-counts boundary dimensionality;  $\chi \times \sigma = 10$  ignores spacetime anchoring;  $D + \chi + \sigma = 11$  adds rather than tensors, violating the product structure of independent subspaces.

The total vacuum rigidity is the trace of these combined structural tensors:

$$\beta_0^{\text{stiff}} = \text{Tr}(T_{\text{anchor}}) + \text{Tr}(T_{\text{pressure}}) = 8 + 3 = \mathbf{11} \quad (4)$$

This derivation is a **universality test**: if the geometric construction rules are truly substrate properties rather than group-theoretic accidents, the same rules must apply to all gauge theories. The QED coefficient  $4/3$  is not fitted; it is the unique ratio of  $2\chi = 4$  (topological load of a virtual pair) to  $D - 1 = 3$  (spatial dilution). Any other value would require either a different topology (violating  $\chi = 2$ ) or different dimensionality (violating  $D = 4$ ).

*b. Topological Distribution: The Screening Term*  
Fermions screen the charge by distributing the boundary topology ( $\chi = 2$ ) across the generation manifold ( $n_{\text{gen}} = 3$ , derived strictly from the Triality of the  $D_4$  internal sector) which is geometrically isomorphic to the available degrees of freedom in the internal color bulk ( $\sigma - \chi = 5 - 2 = 3$ ). Distributing the boundary evenly across these internal dimensions yields the exact fractional screening per flavor:

$$\beta_0^{\text{screen}} = \left( \frac{\chi}{n_{\text{gen}}} \right) \cdot n_f = \frac{2}{3} n_f \quad (5)$$

Combining Rigidity and Screening, we recover the exact QCD Beta Function:

$$\beta_0 = (D\chi) + (\sigma - \chi) - \frac{\chi}{n_{\text{gen}}} n_f = 11 - \frac{2}{3} n_f \quad (6)$$

### Physical Consequence (Asymptotic Freedom):

For  $n_f \leq 16$ , we have  $\beta_0 > 0$ , meaning  $\alpha_s$  decreases logarithmically at high energy, a phenomenon called *asymptotic freedom*, analogous to **negative feedback** in control systems. This emerges because Geometric Stiffness

(11) exceeds Topological Screening ( $2n_f/3$ ) for all Standard Model quark flavors. The strong force weakens at short distances because the vacuum's geometric rigidity dominates the fermion screening contribution.

### 1. Validation 2: Geometric Scaling (Running to $M_\tau$ )

To verify the Stiffness derivation ( $\beta_0$ ), we trace the coupling evolution across energy scales. In the lattice framework, energy scale corresponds to inverse resolution ( $\mu \sim 1/r$ ). The evolution of the coupling is governed by the **Lattice Scaling Law**, where the logarithmic term  $\ln(M_\tau/M_Z)$  represents the number of **Scale Octaves** (doublings) between the two resolutions.

$$\alpha_s(M_\tau) = \frac{\alpha_s(M_Z)}{1 + \frac{\beta_0}{2\pi} \alpha_s(M_Z) \ln(M_\tau/M_Z)} \quad (7)$$

This form, functionally identical to the perturbative Renormalization Group Equation (RGE), emerges here as the linear response of the lattice stiffness ( $\beta_0$ ) to scale transformations.

*Note on Active Degrees of Freedom:* Standard perturbative QCD patches the RGE across quark mass thresholds. In the  $E_8$ -Persistence theory, the global lattice stiffness  $\beta_0$  is a topological invariant of the bulk, driven by the  $n_{gen} = 3$  generation manifold. The geometric scaling therefore utilizes the structural base state ( $n_f = 3 \implies \beta_0 = 9$ ) across the entire macroscopic resolution interval.

Substituting the values:

$$\alpha_s^{(1\text{-loop})}(M_\tau) \approx \frac{0.1179}{1 + \frac{9}{2\pi}(0.1179)(-3.93)} \approx 0.352030151 \quad (8)$$

### 2. The Finite Capacity Correction (The Sterile Channel)

A continuous field theory implicitly assumes infinite information capacity, allowing channels to operate entirely independently regardless of signal density. In the  $E_8$ -Persistence theory, the channel capacity is strictly finite ( $\nu = 16$ ).

As the Saturation Ratio  $\alpha_s$  approaches maximum throughput ( $\gtrsim 0.3$ ), the lattice becomes **capacity-limited**, analogous to a digital communication channel approaching maximum bandwidth. The linear one-loop prediction ( $\alpha_s^{(1\text{-loop})} \approx 0.352030151$ ) fails to account for this hardware limit. We must apply the geometric mechanism of capacity saturation:

**The Geometric Mechanism:** In a discrete lattice with  $\nu$  chiral channels, the coupling  $\alpha_s$  represents the fraction of channels actively transmitting color charge. In the continuum limit (weak coupling), channels operate independently. However, at saturation:

1. **Unitarity Constraint:** Conservation of probability imposes  $\sum_{i=1}^{\nu} p_i = 1$ , which strictly locks exactly one degree of freedom.
2. **Capacity Saturation:** When all  $\nu$  channels are driven to capacity, the system cannot distinguish between “signal” and “saturation noise.” To maintain coherent transmission, the locked channel must remain partially open as a **reference state** (analogous to a pilot tone in dense signaling).
3. **Effective Coupling:** Because of this reference lock, the maximum active signal capacity is limited to exactly  $\nu - 1$  independent channels, while the geometric impedance ( $\alpha^{-1}$ ) is normalized to the full capacity  $\nu$ .
4. **The Sterile Channel ( $\nu_R$ ):** The  $E_8 \rightarrow D_4 \oplus D_4$  decomposition assigns the 16 physical spinors to the  $8_s$  representation of each  $D_4$  factor.  $D_4$  triality is the  $S_3$  symmetry permuting the three inequivalent 8-dimensional representations  $\{8_v, 8_s, 8_c\}$ . Under the  $SU(2)$  action of the topological boundary  $\chi = 2$  (the weak isospin generator), the  $8_s$  decomposes into  $SU(2)$  doublets plus exactly one  $SU(2)$  singlet. This singlet is triality-forced: it corresponds to the direction in  $8_s$  that maps to  $8_c$  under the triality automorphism and therefore carries zero weak isospin — it is the unique  $D_4$  representation fixed by the triality action relative to the  $SU(2)$  boundary. Triality does not merely *identify* this channel after the fact; it *guarantees* exactly one such singlet exists among the 16 chiral states (full derivation deferred to the companion paper on fermion structure). This singlet is the **geometric vacancy**: a topological absence, not a propagating particle. At full saturation, the unitarity constraint locks exactly this channel. The effective active capacity is therefore  $(\nu - 1)/\nu = 15/16$ .

*Why multiplicative?* In Shannon's channel capacity model, a coupling constant represents the fraction of the substrate's transmission probability allocated to a given interaction. If one of  $\nu$  independent channels is blocked (locked as a reference state), the remaining  $(\nu - 1)$  channels transmit freely, but the total available probability space has not changed — it is still normalized to  $\nu$  channels. The coupling therefore scales as the ratio of active-to-total probability:  $(\nu - 1)/\nu$ . An additive correction would imply the blocked channel contributes negative coupling, which has no geometric meaning; the multiplicative ratio correctly reflects the fractional reduction in available transmission probability.

The effective coupling at saturation becomes:

$$\alpha_s^{(\text{eff})} = \alpha_s^{(1\text{-loop})} \cdot \frac{\text{Active Channels}}{\text{Total Capacity}} = \alpha_s^{(1\text{-loop})} \cdot \frac{\nu - 1}{\nu} \quad (9)$$

Numerically:

$$\alpha_s^{(\text{eff})}(M_\tau) = 0.352030151 \times \frac{15}{16} = \mathbf{0.330028266} \quad (10)$$

**Prediction:** 0.330028266

**Experiment:**  $0.330 \pm 0.014$  [10]

**Accuracy:**  $0.00\sigma$

**Physical Interpretation:** This correction is uniquely determined by the chiral capacity  $\nu = 16$ . The “sterile channel” is the geometric vacancy required to maintain unitarity when all active channels saturate; it need not correspond to a propagating degree of freedom in the effective field theory. Standard QFT recovers this effect through higher-order loop diagrams ( $\alpha_s^2, \alpha_s^3, \dots$ ). We obtain the same result directly via the geometric capacity limit, demonstrating that infinite perturbative loop expansions are the continuous approximations of a strict, finite geometric bound.

**Prediction:** This finite-capacity correction should be:

- Negligible in the perturbative regime ( $\alpha_s \lesssim 0.2$ ).
- Significant at strong coupling ( $\alpha_s \gtrsim 0.3$ ).
- Observable in lattice QCD simulations evaluated at finite volume.

### 3. Derivation 3: Geometric Universality — The QED Beta Coefficient

Having validated the QCD stiffness coefficients through running coupling predictions, we now test whether these geometric construction rules extend beyond the Strong Force. If the beta function coefficients are truly geometric properties rather than group-theoretic accidents, the same rules must apply universally to all gauge theories. We test this by deriving the QED coefficient from identical geometric principles.

The QED Beta function coefficient arises from the polarization of the vacuum by virtual pairs:

1. **Topological Load ( $2\chi$ ):** A virtual particle-antiparticle pair creates two topological boundaries ( $\chi = 2 \times 2 = 4$ ).
2. **Spatial Dilution ( $D - 1$ ):** The electric flux distributes over the 3-dimensional spatial volume.

The QED Beta coefficient is the exact ratio of topology to spatial dilution, perfectly matching the standard one-loop result:

$$\beta_0^{\text{QED}} = \frac{2\chi}{D-1} = \frac{4}{3} \quad (11)$$

### 4. Consistency Check: The Z-Pole Running Coupling (Empirical Anchor)

**Numerical Validation (The Unitary Resonance):** To test the QED screening effect at the Z-Pole ( $M_Z$ ), we temporarily adopt the experimentally evaluated vacuum polarization “Screening Fog”  $\Delta\alpha_{\text{total}} \approx 0.0590$  (PDG 2024) as an external input. We utilize this

empirical anchor strictly to isolate and validate the geometric projection overhead ( $\eta$ ) at the Z-Pole. (The *ab initio* derivation of the full fermion spectrum required to calculate  $\Delta\alpha_{\text{total}}$  purely from geometry is the subject of a companion paper, ensuring this temporary anchor does not introduce a free parameter into the fundamental architecture).

Applying this shift to the static vacuum impedance yields a standard effective coupling of  $\alpha_{QFT}^{-1} \approx 137.036 - 8.085 = 128.951$  (where the absolute shift is  $\alpha_{geo}^{-1} \cdot \Delta\alpha_{\text{total}} \approx 8.085$ ). This leaves a  $\approx 1.0$  discrepancy against the experimental value of 127.955.

The  $E_8$ -Persistence theory explicitly resolves this gap. At the exact energy of the Z-Pole ( $M_Z$ ), the vacuum undergoes a **Resonant Transition**, coupling to the Scalar Ground State ( $\Delta^0 = 1$ ). This unitary geometric step is projected onto the physical manifold and is therefore subjected to The Coordinate Overhead ( $\eta$ ). The true geometric impedance is:

$$\alpha^{-1}(M_Z) = [\alpha_{geo}^{-1} - \text{Screening}] - \Delta^0 \cdot \eta \quad (12)$$

$$\alpha^{-1}(M_Z) \approx (137.035999212 - 8.085) - 0.994 \approx \mathbf{127.956689212} \quad (13)$$

**Prediction:** 127.956689212

**Experiment:**  $127.955 \pm 0.010$  [10]

**Accuracy:**  $0.17\sigma$

**Physical Interpretation:** The difference between the low-energy coupling ( $\sim 137$ ) and the Z-pole coupling ( $\sim 128$ ) is the sum of standard fermion screening (QFT) and exactly one unit of geometric resonance ( $\Delta^0 = 1$ ), corrected for manifold projection efficiency ( $\eta$ ).

**Empirical Anchor Declaration:** To isolate the geometric projection overhead  $\eta$  at the Z-pole from the fermion screening contribution, we utilize the experimentally evaluated vacuum polarization  $\Delta\alpha_{\text{total}} \approx 0.0590$  (PDG 2024) as a known boundary condition. This is analogous to using the refractive index of air when verifying Snell’s law: the geometric derivation of  $\eta$  is independent of the medium’s dispersion, but its experimental isolation requires separating the two effects. The *ab initio* derivation of  $\Delta\alpha_{\text{total}}$  from the fermion spectrum is the subject of a companion paper; here, we verify that the geometric correction  $\Delta^0 \cdot \eta$  closes the remaining  $\sim 1.0$  discrepancy once the screening contribution is accounted for.

### D. The Spacetime Partition (Weak Mixing Angle $\sin^2 \theta_W$ )

**Architectural Requirement:** Under Rule 1 (Capacity Partitioning), a persistent system must continuously advance its state into the future. However, a local node

is a heavy informational object. To update the node, the substrate must advance the total structural footprint of the system. It must strictly partition its operational bandwidth between the pure causal “clock tick” (Time) and the structural maintenance of the “hardware” (Space, Payload, and Wires).

**The Geometric Primitives:** The total structural footprint of a node ( $N_{sys}$ ) is the linear sum of its three orthogonal hardware components:

1. **The Address Space ( $R_M$ ):** The spacetime coordinate grid. The fundamental resonance ( $\Delta = 43$ ) distributed across the  $D = 4$  dimensions yields a manifold resolution of  $R_M = 172$  coordinate slots.
2. **The State Payload ( $\nu \cdot \eta$ ):** The active chiral data ( $\nu = 16$ ) occupying the grid, strictly subject to the active transmission overhead ( $\eta$ ), yielding  $\approx 15.9$  active payload states.
3. **The Interaction Bus ( $\sigma$ ):** The  $\sigma = 5$  independent generators required to physically connect the node to its neighbors.

Summing these orthogonal components yields the Total System Footprint:

$$N_{sys} = \underbrace{R_M}_{\text{Address}} + \underbrace{\nu \cdot \eta}_{\text{State}} + \underbrace{\sigma}_{\text{Interaction}} \approx 192.906976744 \quad (14)$$

To maintain causal coherence, a portion of the bandwidth is allocated purely to the temporal update stream ( $\Delta = 43$ , the 1-dimensional causal period) against the total structural footprint of the system ( $N_{sys}$ ):

$$\begin{aligned} \text{Clock Ratio} &= \frac{\text{Temporal Updates}}{\text{Total System Footprint}} = \frac{\Delta}{N_{sys}} \\ &= \frac{43}{192.906976744} \approx \mathbf{0.222905365} \end{aligned} \quad (15)$$

#### 1. Physical Consequence: Electroweak Symmetry Breaking

This geometric bandwidth ratio identically matches the Weak Mixing Angle ( $\sin^2 \theta_W$ ) of the Standard Model. Rather than an arbitrary group-theoretic mixing parameter, it is the structural limit that dictates the physical bifurcation of the Electroweak force:

- **The Photon ( $\gamma$ ):** Represents the  $\Delta$  numerator. Because it couples strictly to the temporal update stream, it carries no structural hardware load. It therefore propagates without geometric impedance as a massless carrier at the speed limit of the substrate ( $c$ ).
- **The W/Z Bosons:** Represent the massive denominator ( $N_{sys} - \Delta$ ). Because they must update the heavy spatial grid, state payload, and interaction bus, their propagation is severely impeded by

the lattice structure, manifesting macroscopically as massive, short-range gauge bosons.

**Prediction:** 0.222 905 365  
**Experiment (Direct Mass):**  $0.222\,91 \pm 0.000\,11$  [10]  
**Experiment (Global Fit):**  $0.223\,54 \pm 0.000\,06$  [10]

**The W-Boson Mass Tension:** The geometric prediction aligns with the **Direct Mass** measurement but deviates by  $-10.58\sigma$  from the Standard Model Global Fit. The global fit derives  $\sin^2 \theta_W$  by propagating correlated constraints across the full electroweak parameter space using continuous perturbative regularization; the geometric prediction arises from a strictly finite, discrete capacity partition. These two procedures need not agree: if the substrate is genuinely discrete, continuous infinite-loop regularization would systematically shift the inferred mixing angle. Whether this  $-10.58\sigma$  tension represents a structural prediction of the discrete substrate or a falsification of the framework depends on the definitive W-mass experimental determination. The direct PDG 2024 measurement ( $M_W = 80.369 \pm 0.013$  GeV) is consistent with the geometric value; the SM global fit remains a genuine challenge this framework must resolve.

*Note on Physical Fields:* The explicit mapping of this Spacetime Partition to the physical gauge fields—demonstrating why the temporal allocation ( $\Delta$ ) manifests as the massless Photon ( $\gamma$ ) and the spatial infrastructure allocation manifests as the massive W/Z bosons—along with the derivation of the absolute link admittances ( $g, g', g_3$ ), is detailed in appendix A.

### E. Jarlskog Invariant $J$ : The Projection Frustration

#### 1. The Architectural Requirement

Under **Rule 2: Reversible Encoding**, the irreducible geometric mismatch of projecting 5-fold symmetry onto a 4-fold manifold cannot be erased; it must be stored as the measurable thermodynamic friction of the causal projection.

The **Toll** pillar of the Capacity Partition is the **Causal Toll**: the irreducible thermodynamic friction of mapping the 5-fold internal symmetry of the lattice ( $\sigma = 5$ ) onto the 4-dimensional spacetime manifold ( $D = 4$ ). In the Standard Model, this manifests as the Jarlskog Invariant ( $J \approx 3 \times 10^{-5}$ ), which quantifies CP violation but whose magnitude has no structural explanation.

We identify  $J$  geometrically as this exact **Projection Frustration**. Time asymmetry arises directly from the interaction between the intrinsic entropic cost of state updates and the geometry of this projection. We define this using three established, parameter-free invariants of the substrate:

These three factors act on geometrically orthogonal subspaces and therefore combine multiplicatively:  $T_{geo}$

operates on channel combinations (combinatorial phase space),  $\phi^2$  on geometric projection (area deficit), and  $\eta$  on bulk propagation (volumetric overhead). By the Entropic Balance Sheet (Paper 1, Section II.B.3), orthogonal contributions to dissipation combine via product, not sum. Any additive combination would violate dimensional consistency: all three factors are dimensionless, but their physical origins are distinct and non-interacting.

1. **The Base Toll ( $T_{geo}$ ):** The entropic cost of a single causal state transition (the Toll pillar) is determined by the Total Node Capacity ( $N = \nu_{\text{Space}} + \nu_{\text{Internal}} = 16 + 16 = 32$ ) and the coupling to the topological boundary:

$$T_{geo} = \frac{1}{N^3} \cdot \frac{\chi}{\sigma} \cdot \left(1 - \frac{\sigma}{D\Delta}\right) \approx 1.185 \times 10^{-5} \quad (16)$$

The cubic dependence ( $1/N^3$ ) counts the minimal three-point vertex configurations; physically, this manifests as the 3-body phase space of Weak decays.

2. **The Projection Frustration ( $\phi^2$ ):** The internal interaction space is characterized by 5-fold symmetry ( $\sigma = 5$ ), which geometrically manifests the non-crystallographic  $H_4$  Coxeter group (the 600-cell). However, a fundamental theorem of crystallography dictates that 5-fold icosahedral symmetries cannot perfectly, periodically tile a 4D discrete lattice without introducing topological defects. When projecting this  $H_4$  geometry onto the periodic 4-dimensional causal manifold, the eigenvalue of the projection matrix strictly defines the geometric area deficit (frustration). This eigenvalue is exactly  $\phi^2 = ((1 + \sqrt{5})/2)^2 \approx 2.618$ , the squared Golden Ratio. This is a rigorous consequence of  $H_4$  Coxeter geometry [11].
3. **The Coordinate Overhead ( $\eta$ ):** Because this frustration signal is a volumetric effect that propagates through the bulk lattice before manifesting at the 4D boundary, it is subject to the Coordinate Overhead ( $\eta$ ).

## 2. The Frustration Formula

The Jarlskog Invariant is the fundamental causal Toll amplified by the Projection Frustration. The frustration acts as a geometric multiplier: every temporal state transition occurs within the constrained 4D projection, and the  $\phi^2$ -scaled area deficit accumulates as additional irreducible entropic cost.

$$\begin{aligned} J_{geo} &= (\phi^2 \cdot \eta) \cdot T_{geo} \\ &= (2.61803 \cdot 0.994186) \cdot 1.1852 \cdot 10^{-5} \\ &\approx 3.08489953 \cdot 10^{-5} \end{aligned} \quad (17)$$

No other combination is permitted:  $T_{geo}$  is the irreducible minimum;  $\phi^2$  is the geometric deficit;  $\eta$  is the bulk propagation correction. The product structure is forced by subspace orthogonality; the numerical value is fixed by the invariant set  $\{\Delta = 43, \nu = 16, \sigma = 5, \chi = 2\}$ .

## 3. Validation: CP Violation

**Prediction:**  $3.08489953 \cdot 10^{-5}$   
**Experiment:**  $(3.08 \pm 0.15) \cdot 10^{-5}$  [10]  
**Accuracy:**  $0.03\sigma$

## 4. Physical Interpretation: Projection Frustration and CP Violation

The CP violation is geometrically analogous to a pentagonal prism casting a shadow: 5-fold symmetry projected onto a 2D surface loses perfect symmetry. The  $E_8$  lattice casts a ‘‘Golden Shadow’’ onto 4D spacetime, the Jarlskog Invariant, the geometric signature of Time’s directional flow.

- **Why CP violation is weak:** The geometric frustration ( $\phi^2 \approx 2.6$ ) amplifies the Toll, but the base transition cost ( $T_{geo} \approx 10^{-5}$ ) is inherently small due to the efficiency of the weak aperture.
- **Why matter dominates:** Without CP violation ( $J = 0$ ), the universe would be perfectly symmetric and annihilate. The non-zero  $J$  creates a topological bias favoring matter over antimatter, which is subsequently amplified by cosmological expansion.
- **The Baryogenesis Gap:** While  $J_{geo}$  matches the Quark Sector mixing ( $J_{CKM}$ ), this magnitude is insufficient to explain the observed baryon asymmetry ( $\eta_B \sim 10^{-9}$ ) through CKM physics alone. The framework structurally requires additional CP violation in the Lepton Sector (PMNS matrix); its geometric derivation is beyond the scope of this paper.

## F. Corollary: The Strong CP Prohibition ( $\theta_{QCD} \equiv 0$ )

While the primary objective of this section is the numerical derivation of the partition coefficients, the geometric isolation of the Capacity Partition yields an immediate, structural resolution to a major open problem in physics: the Strong CP problem. We briefly outline this topological prohibition here, as it demonstrates the predictive power of the partitioned architecture.

In standard QFT, the QCD Lagrangian permits a CP-violating topological term,  $g^2 \theta_{QCD} / (32\pi^2) G\tilde{G}$ , where  $\theta$  is an arbitrary angular parameter. Experimental limits

on the neutron electric dipole moment (nEDM), however, constrain  $|\theta_{QCD}| < 10^{-10}$ . Standard physics offers no structural mechanism for this extreme fine-tuning, motivating *ad hoc* additions like dynamical relaxation via the Peccei-Quinn mechanism and the axion.

The  $E_8$ -Persistence theory resolves the Strong CP problem by **Topological Prohibition**. The interaction subspace that hosts the Strong Force ( $\sigma - \chi = 3$ ) is geometrically orthogonal to the spacetime boundary ( $\chi = 2$ ) where topological winding numbers are defined. This prohibition is enforced by two independent and mutually reinforcing information-theoretic principles:

### 1. Argument 1: The Entropic Ground State Constraint

A non-zero  $\theta_{QCD}$  requires the gauge field to possess a topological winding number around the spacetime manifold ( $\pi_3(S^3)$ ). On a discrete, finite-capacity substrate, this imposes two prohibitive entropic costs:

- 1. Specification Cost:** Because the  $E_8$  substrate is discrete, any non-zero phase angle must manifest as a specific rational geometric ratio of the substrate's integer invariants (as demonstrated for  $\sin^2 \theta_W$  in section IV D). Encoding an arbitrary, continuous parameter  $\theta \in [0, 2\pi)$  would require infinite precision, carrying an unbounded Shannon entropy that the finite node capacity ( $\nu = 16$ ) cannot physically support.
- 2. Coupling Cost:** For a discrete winding number to be meaningful, the color field (residing strictly in the internal  $\sigma - \chi = 3$  subspace) would have to maintain a persistent correlation with the spacetime boundary ( $\chi = 2$  surface). Because these sectors are geometrically orthogonal, this cross-sector coupling requires continuous, irreducible informational overhead to maintain against vacuum decoherence.

The decoupled state where  $\theta_{QCD}$  is geometrically undefined entirely avoids both entropic costs ( $S_{spec} = 0$ ,  $S_{couple} = 0$ ). By the Principle of Least Entropic Action, the vacuum strictly relaxes to the state of minimal information content, the geometric ground state where the winding number simply does not exist:

$$\theta_{QCD} \equiv 0 \quad (\text{Entropic Ground State}) \quad (18)$$

### 2. Argument 2: The Unitary CP Budget and Lorentzian Projection

The lattice's finite capacity imposes a strict conservation law on time-reversal symmetry breaking, quantified by the Jarlskog invariant,  $J$  (which serves as the **Toll** pillar of the Capacity Partition, derived in section IV E).

The  $E_8 \rightarrow D_4 \oplus D_4$  projection assigns the Lorentzian metric signature (and thus the active temporal generator

$\partial_t$ ) exclusively to the spacetime sector. Consequently, the geometric structure allocates the entire CP violation capacity to the chiral spacetime boundary ( $\chi = 2$ ), where it manifests as the CKM phase ( $J \approx 3 \times 10^{-5}$ ), consuming the available temporal budget (as quantified by the Jarlskog invariant derived in section IV E).

The internal symmetry space ( $\sigma - \chi = 3$ ) that hosts the Strong Force is strictly Euclidean; it lacks a projection onto the temporal boundary. The color sector is therefore **informationally isolated from the Arrow of Time**. Because it possesses no temporal generator, its contribution to the CP budget must be identically zero:

$$J_{strong} \equiv 0 \quad \implies \quad \theta_{QCD} = 0 \quad (19)$$

These two arguments are logically independent: either the infinite-precision specification cost (Argument 1) or the geometric isolation of the color sector from temporal generators (Argument 2) would individually prohibit a non-zero  $\theta_{QCD}$ . Their convergence from distinct architectural pillars (Toll and Protocol/Map, respectively) demonstrates that the prohibition is overdetermined by the substrate geometry.

### 3. Comparison to the Axion Mechanism

This geometric resolution is fundamentally different from the dynamical relaxation proposed by the Peccei-Quinn mechanism. The two solutions are mutually exclusive, as summarized in table I.

Table I. Strong CP: Axion vs. Geometric Prohibition

| Feature              | Peccei-Quinn        | $E_8$ -Persistence                 |
|----------------------|---------------------|------------------------------------|
| Mechanism            | Dynamic relaxation  | Topological prohibition            |
| New Particle         | Yes (the Axion)     | No                                 |
| New Symmetry         | Yes ( $U(1)_{PQ}$ ) | No                                 |
| $\theta_{QCD}$ value | Dynamically driven  | Geometrically fixed ( $\equiv 0$ ) |
| Testable via         | Axion search        | Null results in axion search       |

### 4. Falsifiable Predictions

This geometric prohibition generates two hard-falsifiable predictions:

- 1. No QCD Axion:** Because  $\theta_{QCD} = 0$  is the geometric ground state, no dynamical relaxation mechanism is needed. The QCD axion, as proposed to solve the Strong CP problem, is therefore predicted not to exist. A confirmed discovery of the QCD axion with its theoretically expected couplings would explicitly falsify this framework. (*Note: This does not exclude other axion-like particles (ALPs) whose existence is not tied to the Peccei-Quinn mechanism.*)

2. **Zero Strong-Sector nEDM:** The contribution to the neutron electric dipole moment from  $\theta_{QCD}$  is exactly zero. The only non-zero contribution arises from the CKM phase (a weak interaction effect), calculated to be  $d_n^{weak} \sim 10^{-32} e \cdot \text{cm}$  [12]. Given the current experimental limit of  $|d_n| < 1.8 \times 10^{-26} e \cdot \text{cm}$  [13], any confirmed measurement of  $d_n > 10^{-30} e \cdot \text{cm}$  would constitute a major deviation from the Standard Model background and would falsify the  $E_8$ -Persistence Theory.

## V. SYSTEM 5: THE RESONANT INTERFACE (THE ELECTROWEAK SCALE)

The vacuum possesses a strictly finite channel capacity ( $\nu = 16$ ) operating at a fundamental temporal resonance ( $\Delta = 43$ ). In physical terms, this finite bandwidth imposes a strict ceiling on the local energy density of the lattice ( $E = \hbar\omega$ ). The introduction of complex Topological Defects into the bulk saturates this core capacity, threatening the vacuum with Causal Aliasing: the routing failure where the substrate can no longer distinguish which channel belongs to which process when the capacity budget  $\nu$  is exceeded, which manifests physically as runaway energy divergence.

To persist, the system must resolve this complexity. While System 4 defined the *fractional ratio* of bandwidth allocated to this sector (e.g.,  $\sin^2 \theta_W$ ), System 5 derives the *absolute magnitude* of that capacity. This absolute ceiling manifests as the Electroweak Scale.

Under **Rule 3: Impedance Matching**, the vacuum must translate information between the deep geometric bulk (the core lattice) and the localized topological boundary (the particle surface) without dissipative reflection loss. To achieve this, the active energy bandwidth of the Weak Interaction must be perfectly matched to the bulk lattice resonance. Here, we derive the exact phenomenological constants of this interface as the partition coefficients of this geometric matching.

Consistent with *Informational Energetics: Entropic Action* [2], we identify the Higgs sector as the mathematical bounding of this state-space capacity. The internal bandwidth of this resonant interface is rigidly partitioned across the six pillars, defining the following physical parameters:

- **Capacity ( $v$ ):** The VEV; the mass baseline allocated to the resonant interface, against which the vacuum potential is normalized.
- **Map ( $Y$ ):** Scalar Hypercharge; the unitary address. It corresponds to the inverse of the lattice Unitary Ground State ( $\Delta^0 = 1$ ).
- **Protocol ( $\lambda$ ):** Self-Coupling; the self-interaction bandwidth allocation.

- **Governor ( $\lambda|\phi|^4$ ):** The Quartic Potential; The geometric capacity wall enforcing the topological boundary ( $\chi = 2$ ).
- **Toll ( $\mu^2$ ):** Instability Factor; the fractional entropic cost of maintaining the broken symmetry state.
- **Margin ( $y_e$ ):** The Electron Yukawa; the minimal resolvable noise floor.

In the Standard Model, the seemingly arbitrary values of this sector constitute the “Hierarchy Problem,” but they are recognized here as the unique set of partition coefficients required to maintain vacuum stability against causal aliasing.

### A. Capacity: The Higgs VEV ( $v$ )

Under **Rule 1: Capacity Partitioning**, the VEV represents the net informational capacity allocated to the Weak Interaction. We derive this by calculating the total available states over the causal boundary and subtracting the structural overhead.

#### 1. The Bare Geometric Floor

To derive the absolute capacity of the Electroweak interface, we must first define the total informational envelope required to sustain a stable topological defect. Under **Rule 1: Capacity Partitioning**, the system must allocate enough bandwidth to account for every state transition within the defect’s causal influence.

The temporal period  $\Delta = 43$  defines the *causal horizon* of a lattice interaction (the distance information can travel before the system resets). Because information propagates isotropically at the speed of light ( $c = 1$ ), this resonance defines a 2-dimensional causal area  $\Delta \times \Delta = \Delta^2$ . (*Scope Defense: The Euler characteristic  $\chi = 2$  uniquely identifies the defect boundary as a closed orientable 2-sphere ( $S^2$ ) — the only compact surface with  $\chi = 2$ .  $S^2$  is intrinsically 2-dimensional: its natural state-slot area is a 2D cross-section, not a  $D$ -dimensional volume. With  $c = 1$ , both the radial and temporal horizons extend to  $\Delta$ , giving area  $\Delta^2$ . Using  $\Delta^D = \Delta^4$  would count the full causal diamond volume rather than the 2-surface boundary — an overcount by two dimensions. The choice of  $\Delta^2$  is forced by the topology of the defect boundary, not assumed.*)

To distinguish a specific particle (a defect) from the vacuum, this area must be bounded by the topological invariant  $\chi = 2$  (the sphere). The total capacity envelope required to resolve the defect’s identity against the background is therefore the product of its topological signature and its resonant area:  $\chi\Delta^2$ . This represents the total number of “state-slots” available to the interface over one cycle.

The physical VEV is the energy density of this interface capacity: the product of three substrate-level quantities — (1) the available state-slots (the net capacity budget after infrastructure overhead), (2) the minimum energy per state ( $m_e$ , the Landauer limit derived below as the structural floor), and (3) the organizational resistance ( $\alpha^{-1}$ , the impedance governing how much energy is required to coherently address those states). From the total state-slot envelope, the substrate must reserve the infrastructure load it must process per cycle: ( $L_{\text{substrate}} = \nu + D \cdot \Delta = 188[2]$ ). Multiplying by the vacuum impedance ( $\alpha^{-1}$ ) and the energy floor ( $m_e$ ) yields the baseline energy density before geometric corrections.

This transmission impedance is a strictly dimensionless ratio which can be mapped into a physical energy scale by anchoring it to the geometric resolution floor of the lattice: the minimum viable bit of mass ( $m_e$ , derived structurally below in Section VE). This establishes the true physical energy baseline of the field *before* it is subjected to the macroscopic topological dilution and thermal noise corrections ( $\delta_{\text{topo}}, \delta_{\text{noise}}$ ).

$$\begin{aligned} v_{\text{geo}} &= (\chi \Delta^2 - L_{\text{substrate}}) \cdot \alpha^{-1} \cdot m_e \\ &= (2 \cdot 43^2 - 188) \cdot 137.035999212 \cdot 0.51099895 \text{ MeV} \\ &\approx 245.788\,633\,501 \text{ GeV} \end{aligned} \quad (20)$$

## 2. The Geometric Corrections

The  $v_{\text{geo}}$  baseline represents an idealized count of state-slots. To find the *physical* energy density, we must account for the reality of the projection: how this energy is distributed in continuous space and how it is resolved against the substrate noise floor. These effects manifest as exact leading-order linear expansions ( $1 \pm x$ ), representing the first-order response of the vacuum to its own geometric constraints.

*a. Topological Dilution (Spatial Correction)* In the continuum limit, the localized charge of a defect is not a discrete point but a field distributed over a spherical gauge geometry ( $4\pi$ ). This “dilutes” the available energy density because the topological boundary ( $\chi = 2$ ) effectively expands the volume the coupling must cover. We define the **Effective Coupling Volume** ( $D_{\text{eff}}$ ) as the manifold rank ( $D = 4$ ) augmented by this boundary-to-volume ratio:

$$D_{\text{eff}} = D + \frac{\chi}{4\pi} \approx 4.15915 \quad (21)$$

*(Scope Defense: This continuous  $4\pi$  correction represents the macroscopic solid angle of the  $\chi = 2$  boundary; it does not alter the integer dimensionality of the substrate. The correction is  $\alpha/D_{\text{eff}}$  rather than  $\alpha/D$  because the effective volume experienced by the coupling is  $D_{\text{eff}}$ , not*

*the bare lattice rank  $D$ .)* This yields the dilution factor:

$$\delta_{\text{topo}} = \left(1 + \frac{\alpha}{D_{\text{eff}}}\right) \quad (22)$$

*b. Thermodynamic Noise (Temporal Correction)* We must also subtract the energy “lost” to the substrate’s resolution floor. Under **Rule 2: Reversible Encoding**, the **Margin Impedance** ( $Z_{\text{MAR}} \approx 2.9077 \cdot 10^{-6}$ ) defines the minimum resolvable energy unit. Because the vacuum supports exactly 3 fermion generations (the available residual degrees of freedom:  $\sigma - \chi = 3$ ), this noise is partitioned equally across these three independent “signal slots.” *(Scope Defense: The divisor ( $\sigma - \chi$ ) = 3 is the internal interaction bulk minus the topological boundary — the number of geometrically independent generation channels. Dividing by  $\nu = 16$  would conflate chiral capacity with generational noise channels, which are orthogonal degrees of freedom.)* To maintain global solvency, the physical VEV must be reduced by this noise amplitude:

$$\delta_{\text{noise}} = \left(1 - \frac{Z_{\text{MAR}}}{\sigma - \chi}\right) \quad (23)$$

## 3. Synthesis: The Physical VEV

The physical Vacuum Expectation Value is the geometric floor modulated by the simultaneous application of these topological and thermodynamic factors.

$$v_{\text{phys}} = v_{\text{geo}} \cdot \delta_{\text{topo}} \cdot \delta_{\text{noise}} \quad (24)$$

Substituting the derived values:

$$\begin{aligned} v_{\text{phys}} &= (245.788\,633\,501 \text{ GeV}) \cdot \\ &\left(1 + \frac{\alpha}{4.159}\right) \cdot \left(1 - \frac{2.9077 \cdot 10^{-6}}{3}\right) \\ &\approx 246.219\,637\,823 \text{ GeV} \end{aligned} \quad (25)$$

**Prediction:** 246.219 637 823 GeV

**Experiment:** 246.219 65  $\pm$  0.000 06 GeV [10]

**Accuracy:**  $-0.20\sigma$

## 4. Output: The Fermi Constant ( $G_F$ )

With the interface capacity  $v_{\text{phys}}$  fixed, a propagating fermion couples to it through a 1D chiral projection. The Fermi Constant  $G_F$  is the inverse squared capacity of the interface, attenuated by this projection factor.

*a. The Geometric Origin of  $\sqrt{2}$*  For  $v_{\text{phys}}$  to mediate a force, it must be projected onto the 1D chiral axis of the fermion’s trajectory. The interface amplitude is isotropically distributed on  $S^1$ ; projecting onto a single axis averages as  $\langle \cos^2 \theta \rangle = 1/2$ , yielding an RMS projection factor of exactly  $1/\sqrt{2}$ .

*b. The Result* The Fermi constant is therefore the inverse squared capacity of the interface, attenuated by this geometric projection factor:

$$G_F = \frac{1}{\sqrt{2}(v_{phys})^2} \approx 1.16637882 \times 10^{-5} \text{ GeV}^{-2} \quad (26)$$

**Prediction:**  $1.16637882 \times 10^{-5} \text{ GeV}^{-2}$

**Experiment:**  $(1.1663788 \pm 0.0000006) \cdot 10^{-5} \text{ GeV}^{-2}$  [10]

**Accuracy:**  $0.04\sigma$

## B. Map: Scalar Hypercharge ( $Y = 1$ )

The Resonant Interface must maintain a unique, resolvable identity within the substrate it regulates. Without a discrete address, the regulator cannot distinguish its own state transitions from background vacuum fluctuations and loses coherence. The **Map** pillar assigns this address.

As established in [2], the lattice’s fundamental coordinate origin is the Unitary Ground State ( $\Delta^0 = 1$ ). For the regulator to maintain continuous resonance with the vacuum without introducing a new, independent degree of freedom, its address must be this same minimal unit charge:

$$Y = \Delta^0 = 1 \quad (27)$$

This is the unique assignment consistent with the Closure Constraint: any other value would require encoding an additional free parameter into the substrate.

Combined with the topological boundary ( $\chi = 2$ ), this unitary hypercharge geometrically necessitates the  $SU(2)$  doublet structure, allowing the regulator to mediate between the chiral matter sector and the gauge field bulk. (*Note: A complete derivation of how all Standard Model fermion hypercharges emerge as exact geometric ratios of the lattice invariants is provided in Appendix B.*)

## C. Protocol: Self-Coupling ( $\lambda$ )

The Resonant Interface must regulate its own amplitude against perturbation. Without a self-regulatory feedback, scalar field fluctuations grow unbounded, causing the interface to decohere. The **Protocol** pillar allocates the bandwidth required for this internal self-regulation. Under **Rule 1: Capacity Partitioning**, this self-coupling  $\lambda$  is the exact fraction of the defect’s total informational complexity dedicated to maintaining internal stability. (The Standard Model labels this the Higgs self-coupling and treats it as a free parameter; here it is a determined geometric fraction.)

## 1. The Intrinsic Structural Load ( $L_{intrinsic}$ )

The “cost” of the defect is a fraction of its total informational complexity. As established in [1], a defect’s total informational complexity is its **Intrinsic Load** ( $L_{intrinsic} = 23$ ): the sum of its orthogonal geometric components—its chiral capacity ( $\nu = 16$ ), its interaction degrees of freedom ( $\sigma = 5$ ), and its topological identity lock ( $\chi = 2$ ).

## 2. The Regulatory Bandwidth

The “budget” available for self-regulation is the capacity remaining in the internal interaction bulk ( $\sigma - \chi = 3$ ) after the identity lock is paid. However, the system must pay a **Resonant Cost** ( $1/\Delta$ ) to maintain temporal synchronization with the lattice. Without reserving this fractional bandwidth to “keep time,” the scalar field would drift out of phase with the fundamental update cycle ( $\Delta = 43$ ), causing the subsystem to decohere and decay.

The net bandwidth available for self-regulation is therefore:

$$\begin{aligned} \text{Net} &= (\sigma - \chi) - \frac{1}{\Delta} \\ &= 3 - \frac{1}{43} \approx 2.976744186 \end{aligned} \quad (28)$$

## 3. The Resulting Coupling

The coupling  $\lambda$  is the ratio of this regulatory budget to the total structural load. It represents the “stiffness” required to anchor the defect against the vacuum’s entropic current:

$$\lambda = \frac{\text{Net}}{L_{intrinsic}} = \frac{2.9767}{23} \approx \mathbf{0.12942366} \quad (29)$$

**Prediction:**  $0.12942366$

**Experiment:**  $0.129 \pm 0.005$  [10]

**Accuracy:**  $0.08\sigma$

## D. Toll: Instability ( $\mu^2$ )

In the Standard Model, the negative mass-squared parameter ( $-\mu^2$ ) is an axiomatic input required to drive Spontaneous Symmetry Breaking. In the  $E_8$ -Persistence theory, this corresponds to the **Toll**: the entropic cost required to maintain the unstable, unbroken symmetric state ( $\phi = 0$ ) against the vacuum.

### 1. The Instability Action

As established by the topological boundary ( $\chi = 2$ ), the Electroweak interface operates as an  $SU(2)$  doublet, requiring two independent complex degrees of freedom. Under **Rule 1: Capacity Partitioning**, each of these degrees of freedom must pay the self-regulation cost ( $\lambda$ ) derived in the previous section. The total instability action ( $T_H$ ) mandated by the substrate is therefore:

$$T_H = \chi \cdot \lambda = 2\lambda \approx 0.259 \quad (30)$$

### 2. Structural Consistency: Recovering $\mu^2 = \lambda v^2$

We validate this entropic cost by calculating the physical mass parameter  $\mu^2$ .  $\mu^2$  is the fractional Toll ( $T_H$ ) scaled across the available capacity ( $v$ ) of the interface. Dividing the total action by the two degrees of freedom yields:

$$\mu^2 = \frac{T_H}{2} v^2 = \frac{2\lambda}{2} v^2 = \lambda v^2 \quad (31)$$

This acts as a rigorous **Consistency Check**. By recovering the standard physical relation ( $\mu^2 = \lambda v^2$ ) purely from geometric invariants, we demonstrate that the “tachyonic” nature of the Higgs potential is the literal manifestation of the entropic Toll driving the system toward a stable Quiescent Equilibrium.

### E. The Resolution Floor: The Electron Yukawa ( $y_e$ )

Under **Rule 2: Reversible Encoding**, any information that cannot be perfectly discretized onto the lattice grid must be stored as structural metadata to avoid dissipative loss (Landauer erasure). This metadata manifests physically as mass. The **Electron Yukawa** ( $y_e$ ) represents the **Margin**: the absolute resolution floor of the vacuum, below which no distinct topological state can be resolved.

#### 1. Derivation: The Diluted Bit

On a discrete substrate, the smallest possible signal is a single unit of action (one bit). To find the entropic cost of this signal, we calculate the dilution of this “Unit Bit” across the total configuration space of a topological defect. As established in [1], this volume is defined by the product of the defect’s address budget ( $L_{embed}$ ), its interaction aperture ( $\sigma + 1$ ), and its causal resonance area ( $\Delta^2$ ):

$$y_{e,bare} \equiv Z_{MAR} = \frac{1}{V_{config}} = \frac{1}{\underbrace{L_{embed}}_{\text{Budget}} \cdot \underbrace{(\sigma + 1)}_{\text{Aperture}} \cdot \underbrace{\Delta^2}_{\text{Area}}} \quad (32)$$

#### 2. Physical Interpretation: The Landauer Limit

Physically, the electron represents the **Landauer Limit** of the vacuum. Any topological knot with a coupling  $y < y_{e,bare}$  would possess a binding energy lower than the substrate’s thermal noise floor ( $E < k_B T_{eff} \ln 2$ ). Such a state would be informationally indistinguishable from vacuum fluctuations and would immediately evaporate. The electron is stable because it is the **Minimum Viable Particle**, the physical instantiation of the smallest non-erasable unit of mass.

#### 3. Geometric Renormalization: The Projection Stress

Standard QFT treats the difference between “bare” and “physical” mass as a result of virtual particle loops. In the  $E_8$ -Persistence framework, this correction is the **Geometric Stress** of embedding a high-dimensional knot into a lower-dimensional manifold.

A topological defect possesses an internal interaction structure of rank  $\sigma = 5$ , but it must persist within a spacetime manifold of rank  $D = 4$ . This mismatch ( $\sigma > D$ ) creates an irreducible “Overpressure” on the surrounding lattice. We define the **Projection Coefficient** ( $\mathcal{P}$ ) as the ratio of this internal load to the manifold’s capacity:

$$\mathcal{P} = \frac{\text{Interaction Rank}}{\text{Manifold Rank}} = \frac{\sigma}{D} = 1.25 \quad (33)$$

The physical effective coupling ( $y_{e,phys}$ ) is the bare floor enhanced by this projection stress acting through the vacuum impedance ( $\alpha$ ):

$$y_{e,phys} = y_{e,bare} \cdot (1 + \mathcal{P} \cdot \alpha) = y_{e,bare} \cdot (1 + 1.25\alpha) \quad (34)$$

Consistent with the dimensional budget of the Entropic Action [2], higher-order feedback terms are suppressed by  $\alpha^2 \approx 5 \times 10^{-5}$  and are safely truncated.

#### 4. The Physical Result

Substituting the invariants ( $L_{embed} = \nu + \sigma + (D + 1)\chi = 31$ ,  $\sigma + 1 = 6$ ,  $\Delta = 43$ ) and the derived  $\alpha$ :

$$y_{e,phys} \approx \left( \frac{1}{31 \cdot 6 \cdot 43^2} \right) \cdot \left( 1 + \frac{1.25}{137.036} \right) \approx 2.93422684 \cdot 10^{-6} \quad (35)$$

**Prediction:**  $2.93422684 \cdot 10^{-6}$   
**Experiment:**  $(2.935 \pm 0.001) \cdot 10^{-6}$  [10]  
**Accuracy:**  $-0.77\sigma$

### 5. Physical Consequences: The Spectral Floor

This geometric derivation explains two fundamental features of the particle spectrum:

- **Electron Stability:** The electron sits exactly at the minimum resolvable mass. There is no lower energy state for it to decay into while still maintaining its topological identity.
- **Neutrino Mass:** Massive neutrinos possess Yukawa couplings  $y_\nu \ll y_e$ . This pushes them below the vacuum's resolution floor, meaning they cannot acquire mass via the standard Higgs mechanism (which terminates at  $y_e$ ). Their mass must arise from a secondary, non-topological mechanism (derived in future work).

### F. Output: The Regulatory Excitation (The Higgs Mass, $m_H$ )

The Higgs mass is the resonant tone of the capacity wall: if the VEV defines the boundary containing the scalar field,  $m_H$  is the frequency that boundary emits under excitation—the geometric stiffness of the capacity bound made physical.

Standard model-building treats the Higgs boson as a fundamental particle akin to quarks or leptons, albeit with scalar properties. In the  $E_8$ -Persistence framework, the Higgs is fundamentally distinct: it is not a Topological Knot (matter), but rather **Geometric Infrastructure**. It is the physical manifestation of the Resonant Interface itself.

If the Vacuum Expectation Value ( $v$ ) represents the total state-space *capacity* allocated to the interface, and the self-coupling ( $\lambda$ ) represents the *stiffness* of that partition, then the Higgs Mass ( $m_H$ ) is the energetic cost of perturbing this regulatory infrastructure. It is the resonant vibration of the boundary.

Because the Higgs is not a free particle but the structural response of the vacuum, its mass is strictly dictated by the closure of the geometric system. By combining the interface capacity ( $v_{phys}$ ) and the regulatory protocol ( $\lambda$ ) derived in the preceding sections, the physical mass is determined by the curvature of the geometric capacity wall. As established in [2], the Higgs potential takes the form  $V(\phi) = \lambda(|\phi|^2 - v^2/2)^2$ . The second derivative of this potential at its minimum  $|\phi| = v/\sqrt{2}$  gives  $m_H^2 = 2\lambda v^2$ , so:

$$\begin{aligned} m_H &= \sqrt{2\lambda} \cdot v_{phys} \\ &\approx 125.269\,263\,77 \text{ GeV} \end{aligned} \quad (36)$$

**Prediction:**  $125.269\,263\,77 \text{ GeV}$   
**Experiment:**  $125.20 \pm 0.11 \text{ GeV}$  [10]  
**Accuracy:**  $0.63\sigma$

**Architectural Conclusion:** This formulation resolves the ontological status of the Higgs boson. It does not exist alongside fermions in a spectral scan of lattice excitations because it is the "stage" rather than the "actor." Its mass is not an arbitrary parameter subject to infinite radiative corrections; it is the mathematically rigid impedance of the vacuum's self-regulatory partition.

### G. Validation: Global Impedance Matching

The structural test of the Resonant Interface is whether it successfully satisfies **Rule 3: Impedance Matching**. For a nested subsystem to persist, its internal regulation must perfectly match the geometric aperture of the signals it governs.

*Application Rule for Coordinate Overhead  $\eta$ :* The Coordinate Overhead  $\eta = 171/172$  is the geometric cost of projecting a bulk lattice quantity onto the lower-dimensional coordinate manifold traversed by propagating fields. The rule is:  $\eta$  applies to any quantity that represents the *aperture as experienced by a propagating signal* (defined in terms of the manifold coordinate grid);  $\eta$  does not apply to *intrinsic geometric impedances* defined within the bulk lattice prior to any propagation. For this impedance match:  $Z_H$  is the self-impedance of the lattice regulator (a bulk invariant, independent of the coordinate projection);  $A_{weak}$  is the interaction aperture as seen by propagating fermions (projected onto the manifold). The  $\eta$  asymmetry is structural, not a modeling choice.

We verify this by comparing the **Higgs Impedance** (the controller) to the **Weak Projected Aperture** (the load).

1. **The Controller ( $Z_H$ ):** We define the Higgs Impedance as the inverse of its regulation protocol ( $1/\lambda$ ). Crucially, because this regulator operates under the Entropic Action filter [2], its effective impedance is exponentially modulated by its action cost ( $T_H = 2\lambda$ ). ( $Z_H$  is the *intrinsic self-impedance of the regulator, defined before propagation and therefore not subject to Coordinate Overhead  $\eta$ , which applies only to signals transmitted over the bulk coordinate grid.*)

$$Z_H = \frac{1}{\lambda} e^{-2\lambda} \approx \frac{1}{0.129424} (0.7719) \approx \mathbf{5.964449406} \quad (37)$$

2. **The Aperture ( $A_{weak}$ ):** The hardware aperture of the Weak interaction is the sum of the lattice primitives involved in a state transition: the interaction links ( $\sigma = 5$ ) plus the node occupancy bit (+1). Because this aperture is projected onto the

physical manifold, it is subject to the **Coordinate Overhead** ( $\eta$ ):

$$A_{weak} = (\sigma + 1) \cdot \eta \approx 6 \cdot 0.994186 \approx \mathbf{5.965116279} \quad (38)$$

*a. The Result* The regulator impedance ( $Z_H \approx 5.9644$ ) matches the projected aperture ( $A_{weak} \approx 5.9651$ ) to within **0.01%**.

This convergence confirms that the **Electroweak Scale** is the unique energy magnitude required to satisfy global impedance matching between the lattice resonance and the surface of matter. The  $O(10^{-4})$  residual is numerically consistent with the  $O(\alpha^2)$  truncation limits of the framework's geometric corrections ( $\alpha^2 \approx 0.5 \times 10^{-4}$ ).

The difference represents the **maximum precision** allowed by a first-order geometric derivation and identifies the theoretical floor of the vacuum's entropic drag. In the IE framework, this residual is the **minimum entropic cost** required for information to cross the interface boundary. This structural lock demonstrates that the Resonant Interface is a mathematically closed, self-sustaining regulatory subsystem, resolving the Hierarchy Problem as a necessary condition for substrate persistence.

## VI. SYSTEM 6: THE GEOMETRIC DEPTH ( $\alpha_G, M_P, \rho_{vac}$ )

### A. The Architectural Requirement

Gravity is traditionally treated as a distinct fundamental force described by General Relativity, operating with a coupling constant  $G$  that is inexplicably  $\sim 10^{40}$  times weaker than the gauge forces (in the conventional proton-mass comparison  $Gm_p^2/e^2$ ), and a vacuum energy  $\sim 10^{120}$  times smaller than QFT predictions. This extreme disparity forces the Planck Mass ( $M_P$ ) and Cosmological Constant ( $\Lambda$ ) to be inserted into the Standard Model as manual scaling limits.

We identify Gravity as the **Geometric Depth** of the lattice. Just as the Higgs regulates the energy density of Matter at the local surface, Gravity is the nested regulatory subsystem that maintains the structural integrity of the bulk geometry. It is an **Attenuated Signal**. Gauge forces originate at the topological boundary (the particle's "surface",  $r = 0$ ), while gravity originates at the lattice centroid (the particle's "core",  $r = \Delta/2$ ). (*This spatial partitioning of force origins is introduced here as a structural consequence of the  $D_4 \oplus D_4$  decomposition; Papers 1 and 2 established the functional forms of these forces but not this loci assignment.*) The apparent gravitational weakness is strictly the signal attenuation across this geometric depth.

As derived in *Informational Energetics: Entropic Action* [2], the Bulk Regulator's architectural pillars mathematically necessitate the Einstein-Hilbert action (a massless spin-2 field preserving diffeomorphism invariance) to

conserve local capacity and maintain volumetric resolution. In this section, we assume those derived functional forms and focus exclusively on calculating their exact numerical partition coefficients using the nested partitioning rules established in Section II: the gravitational coupling ( $\alpha_G$ ), the Planck Mass ( $M_P$ ), and the Cosmological Constant ( $\Lambda$ ).

To satisfy **Rule 3: Impedance Matching**, the internal bandwidth of this bulk regulator is rigidly partitioned using the lattice invariants:

- **Capacity** ( $M_P$ ): The Unity Threshold where the bulk geometric impedance equals 1.
- **Map** ( $h_{\mu\nu}$ ): The Traceless Transverse metric perturbation.
- **Protocol** ( $\alpha_G$ ): The core-to-surface attenuation efficiency.
- **Governor (Diffeomorphism Invariance)**: The local capacity conservation strain.
- **Toll** ( $M_P^2(\partial h)^2$ ): The causal update cost enforcing the light-cone limit.
- **Margin** ( $\rho_{vac}$ ): The bulk resolution floor (Vacuum Energy).

### B. Protocol: The Gravitational Coupling $\alpha_G$

To maintain persistence under **Rule 1: Capacity Partitioning**, the Bulk Protocol must propagate on the **Residual Capacity**, the bandwidth strictly left over after the primary surface gauge allocations are filled (the global solvency of this partition is verified in section VII).

#### 1. Residual Capacity ( $B_{res}$ )

The available bandwidth for gravity is the Total Chiral Capacity ( $\nu$ ) minus the allocations for Topological Storage and Surface Gauge Load.

$$B_{res} = \nu - \underbrace{\frac{\chi}{\sigma - \chi}}_{\text{Boundary Storage}} - \underbrace{\alpha}_{\text{Gauge Load}} \quad (39)$$

- **Boundary Storage** ( $\frac{2}{3} \approx 0.67$ ): The capacity consumed by topological charge quantization. Geometrically, this is the strict ratio of the topological boundary ( $\chi = 2$ ) to the internal interaction bulk ( $\sigma - \chi = 3$ ), representing the informational overhead required to map the 2D surface boundary onto the 3D internal bulk generators.
- **Gauge Load** ( $\approx 0.007$ ): The bandwidth reserved for Electromagnetic synchronization.

Substituting the invariants ( $\nu = 16, \sigma = 5, \chi = 2$ ):

$$B_{res} \approx 16 - \frac{2}{3} - 0.00729 \approx \mathbf{15.326035981} \quad (40)$$

## 2. Harmonic Attenuation (The Lattice Depth)

The gravitational signal must propagate from the lattice core ( $r = \Delta/2$ ) to the surface ( $r = 0$ ). Each lattice layer is a discrete information-processing step: a signal quantum is either coherently relayed to the next layer (with probability  $(1 - \alpha)$ ) or lost to vacuum fluctuations (with probability  $\alpha$ ). As established in Paper 1 [1],  $\alpha$  is the per-bond fidelity loss of the  $E_8$  lattice — the fraction of a transmitted state that fails to propagate coherently across one lattice link. This is the same  $\alpha$  that governs the electromagnetic vacuum impedance ( $\alpha^{-1}$ ): both are the same geometric quantity, one viewed as coupling strength at the surface, the other as attenuation rate through the bulk. After  $r$  independent hops, the surviving signal fraction is  $(1 - \alpha)^r = \alpha^r$  (in the small- $\alpha$  limit; see Paper 1 for the exact  $E_8$  lattice derivation). The Beer-Lambert law is the macroscopic continuous limit of this discrete per-hop calculation.

The origin of this bulk signal is strictly dictated by the foundational symmetry of the lattice. For the  $E_8 \rightarrow D_4 \oplus D_4$  projection to be persistent, it must preserve the self-duality of the parent lattice. This requires the bulk resonance to treat both orthogonal  $D_4$  sublattices perfectly symmetrically. Any excitation centered on an integer-valued lattice node would inherently break this symmetry by selecting one sublattice over the other.

Therefore, the bulk resonance must manifest as a continuous standing wave,  $\Psi(n) = \cos(2\pi n/\Delta)$ , spanning the causal loop. (*Scope Defense: The cosine is the unique fundamental mode on a closed loop of period  $\Delta$  with a single antinode at the center — minimal nodal structure consistent with the self-duality requirement. Higher harmonics are suppressed because their additional nodes break the  $D_4$  symmetry.*) To preserve exact self-duality, the geometric origin (the point of maximal bulk energy density) must be located exactly at the continuous geometric midpoint, the **Centroid** of the wavelength. For a causal period of  $\Delta = 43$ , this uniquely fixes the origin at:

$$r_{core} = \frac{\Delta}{2} = 21.5 \quad (41)$$

The **Lattice Depth** ( $r = 21.5$ ) sets the bulk attenuation scale. The attenuation follows an exponential decay law (analogous to the Beer-Lambert law):

$$\text{Attenuation Factor} = \alpha^{\text{Radius}} = \alpha^{21.5} \approx \mathbf{1.143 \times 10^{-46}} \quad (42)$$

## 3. The Gravitational Coupling Calculation

The coupling is the product of the spare capacity (Bandwidth) and the attenuation factor (Depth). This represents the **Maximal Efficient Coupling** possible without disrupting the surface fields.

$$\alpha_G = B_{res} \cdot \alpha^{\Delta/2} \quad (43)$$

$$\alpha_G = 15.326 \times (1.143 \times 10^{-46}) \approx \mathbf{1.75179958 \cdot 10^{-45}} \quad (44)$$

**Prediction:**  $1.75179958 \cdot 10^{-45}$

**Experiment:**  $(1.752 \pm 0.001) \cdot 10^{-45}$  [14]

**Accuracy:**  $-0.20\sigma$

**Physical Interpretation:** Gravity appears  $\sim 10^{40}$  times weaker than electromagnetism in the proton-mass comparison ( $Gm_p^2/e^2$ ). The electron-normalized coupling derived here is  $\alpha_G \approx 1.75 \times 10^{-45}$ , reflecting the same geometric depth of 21.5 lattice layers but normalized to  $m_e$  rather than  $m_p$ . In both cases, the apparent weakness is a consequence of the centroid attenuation  $\alpha^{\Delta/2}$ , not a fundamental disparity.

## C. Capacity: The Planck Mass $M_P$ (The Unity Threshold)

Under **Rule 3: Impedance Matching**, the Planck Mass is the unique energy scale where the bulk geometric impedance achieves unity, enabling lossless signal transmission across the full lattice depth.

In the  $E_8$ -Persistence theory, the Planck Mass is the **Unity Threshold**. It is the mass scale at which the sheer magnitude of the signal exactly compensates for the geometric attenuation, allowing the core to couple to the surface with unit strength ( $\alpha_G \rightarrow 1$ ).

It connects the natural unit of the Surface (the Electron,  $m_e$ ) to the natural unit of the Core (the Planck Mass). In standard natural units with the non-reduced Planck mass convention ( $M_P = \sqrt{\hbar c/G} \approx 1.22 \times 10^{19}$  GeV), the dimensionless gravitational coupling between two electrons is  $\alpha_G = Gm_e^2 = (m_e/M_P)^2$ . Therefore, the mass ratio is strictly the inverse square root of the coupling:

$$M_P = m_e \cdot \frac{1}{\sqrt{\alpha_G}} = m_e \cdot \frac{1}{\sqrt{B_{res} \cdot \alpha^{\Delta/2}}} \quad (45)$$

Expanding the term reveals the direct parameter-free dependence on the Fine-Structure Constant and the Heegner Resonance (the Stark-Heegner number  $\Delta = 43$  derived as the fundamental lattice period in Paper 1 [1]):

$$M_P = \frac{m_e}{\sqrt{B_{res}}} \cdot \alpha^{-\Delta/4} \quad (46)$$

### 1. Numerical Result

$$M_P = \frac{0.51099895 \text{ MeV}}{\sqrt{15.326}} \cdot (137.035999212)^{\Delta/4} \quad (47)$$

$$M_P \approx 1.22089355 \times 10^{19} \text{ GeV} \quad (48)$$

**Prediction:**  $1.220\,893\,55 \times 10^{19}$  GeV  
**Experiment:**  $(1.22091 \pm 0.00001) \cdot 10^{19}$  GeV [14]  
**Accuracy:**  $-1.65\sigma$

#### D. Margin: The Vacuum Energy $\rho_{vac}$ (Lattice Resolution Floor)

Under **Rule 1: Capacity Partitioning**, the vacuum energy density is set by the thermal noise floor of the discrete substrate, the minimum resolvable bulk vibration after all higher-energy modes are allocated to the gauge and matter sectors.

Standard Quantum Field Theory predicts a vacuum energy density  $\rho_{vac} \sim M_P^4$ , summing zero-point fluctuations up to the Planck scale. This yields the “Vacuum Catastrophe”: a prediction  $\sim 10^{120}$  times larger than cosmological observations (the precise factor depends on normalization convention; this framework recovers  $\sim 10^{123}$  from the ratio  $(M_P/M_{min})^4$ ).

In the  $E_8$ -Persistence theory, we distinguish between two distinct resolution floors:

- **The Matter Floor** ( $m_e \approx 0.511$  MeV): The lightest stable *Topological Knot*, derived in System 5 as the Margin for chiral states.
- **The Lattice Floor** ( $M_{min}$ ): The minimum resolvable *Bulk Vibration* (Phonon), set by the thermal noise limit of the discrete substrate.

While stable matter requires topological closure ( $\chi = 2$ ), bulk vibrations are constrained only by the **Information-Theoretic Noise Floor** of the lattice.

##### 1. The Thermal Resolution Limit

The minimum resolvable energy  $M_{min}$  is determined by the Nyquist-Shannon sampling limit applied to the lattice phase space. The lowest recoverable frequency corresponds to a single quantum of thermal action distributed across the maximum available coordinate volume:

$$M_{min} = m_e \cdot \frac{\mathcal{A}_{thermal}}{\mathcal{V}_{modes}} \quad (49)$$

where:

- **Thermal Coupling** ( $\mathcal{A}_{thermal} = \pi\alpha$ ): The exact geometric phase cross-section ( $\pi$ ) modulated by the linear vacuum admittance ( $\alpha$ ). (*Scope Defense: Unlike stable matter, which requires a 2D spherical topological boundary ( $\chi = 2$ ) to anchor itself, transient thermal fluctuations are unanchored. They couple to the bulk strictly through the minimal 1D phase boundary: the Wilson Loop on a unit circle has circumference  $\pi$  in the half-period parameterization, representing the minimal distinguishable phase arc on a compact  $U(1)$  manifold. Because these are minimal perturbative excitations,*

*they couple linearly through the bare transmission probability ( $\alpha^1$ ), representing the exact geometric fraction of thermal energy that can excite a bulk phonon).*

- **Mode Density** ( $\mathcal{V}_{modes} = \nu\Delta^3$ ): The total number of orthogonal phase space cells, given by the chiral capacity ( $\nu = 16$ ) distributed across the 3-dimensional spatial resonant volume ( $\Delta^3 = 43^3$ ). (*Scope Defense: Spatial volume  $\Delta^3$  is used instead of spacetime volume  $\Delta^4$  because thermal equilibrium is rigorously defined across a spacelike hypersurface of the causal manifold.*)

Thus, the ground state frequency is:

$$M_{min} = m_e \cdot \frac{\pi\alpha}{\nu\Delta^3} \approx 9.2 \times 10^{-12} \text{ GeV} \quad (50)$$

This energy scale corresponds to a characteristic wavelength:

$$\lambda_{min} = \frac{1}{M_{min}} \approx 21 \mu\text{m} \quad (51)$$

This places the lattice noise floor in the mid-infrared range, comparable to molecular vibrational scales. Whether this numerology reflects a structural connection between the vacuum floor and macroscopic thermodynamic phenomena is left as an open question.

##### 2. The Vacuum Energy Density

The cosmological constant is the zero-point energy density of oscillations at this minimum energy scale:

$$\rho_{vac} = \frac{\alpha}{2} M_{min}^4 \quad (52)$$

where the factor  $\alpha/2$  arises strictly from:

- $1/2$ : The zero-point energy averaging ( $\langle E \rangle = \frac{1}{2}\hbar\omega$ ).
- $\alpha$ : The vacuum admittance (the fraction of modes thermally accessible to the boundary).

Normalizing to the Planck density:

$$\frac{\rho_{vac}}{M_P^4} = \frac{\alpha}{2} \left( \frac{M_{min}}{M_P} \right)^4 \approx 1.18103720 \cdot 10^{-123} \quad (53)$$

**Prediction:**  $1.18103720 \cdot 10^{-123}$   
**Experiment:**  $(1.38 \pm 0.20) \cdot 10^{-123}$  [10]  
**Accuracy:**  $-0.99\sigma$

The Cosmological Constant is the strict volumetric resolution floor of the finite lattice. By enforcing a physical channel capacity limit, the Vacuum Catastrophe is resolved: with no substrate modes above  $M_{min}$ , the vacuum energy density naturally falls to the observed dark energy scale without cancellation or fine-tuning. The conventional factor of  $\sim 10^{120}$  ( $10^{123}$  in the normalization of this calculation) is the ratio  $(M_P/M_{min})^4$  — purely geometric.

### 3. Physical Interpretation

The  $10^{120}$  discrepancy arises entirely from different assumptions about the vacuum structure:

- **Standard QFT:** Assumes the vacuum is a continuous manifold fluctuating at all scales up to  $M_P$ , yielding  $\rho_{QFT} \sim M_P^4$ .
- **$E_8$ -Persistence:** Treats the vacuum as a finite-bandwidth information channel. The finite channel capacity ( $\nu = 16$ ) imposes a hard UV ceiling at  $M_{min}$ : modes above this scale have no available bit-address in the substrate and are structurally excluded. Only the thermal noise floor survives, yielding  $\rho_{E8} \sim M_{min}^4$ .

The ratio is purely geometric:

$$\frac{\rho_{QFT}}{\rho_{E8}} \sim \left( \frac{M_P}{M_{min}} \right)^4 \sim \left( \frac{\nu \Delta^3}{\pi \alpha} \right)^4 \sim 10^{123} \quad (54)$$

### E. Consistency Check: Bulk Impedance Closure

To confirm gravity is a mathematically closed persistent subsystem, we verify that the network equations close flawlessly at the Planck scale. While the Higgs impedance matching (System 5) tested independent geometric quantities against each other, the gravitational scale  $M_P$  is explicitly defined by the unity coupling condition. Therefore, this serves as a rigorous **algebraic consistency check** rather than an independent validation: the defined parameters must naturally close the network equations without generating residual scalar mismatches.

The gravitational impedance at the Planck scale must satisfy the unity condition:

$$Z_G(M_P) = \frac{m_e}{M_P \sqrt{\alpha_G}} = \frac{m_e}{M_P \sqrt{B_{res} \cdot \alpha^{\Delta/2}}} \equiv 1 \quad (55)$$

This closes by definition: since  $M_P \equiv m_e / \sqrt{\alpha_G}$ , we have  $m_e / M_P = \sqrt{\alpha_G}$ , giving  $Z_G = \sqrt{\alpha_G} / \sqrt{\alpha_G} = 1$ . The closure is algebraic rather than predictive — it confirms internal consistency: the Planck Mass is the unique scale at which the geometric attenuation  $\sqrt{\alpha_G}$  is exactly compensated by the mass ratio  $m_e / M_P$ . Physically, this means signals traverse the full 21.5-layer lattice depth with unit coupling efficiency at exactly  $M_P$ . At energies below  $M_P$ , gravity is geometrically attenuated ( $Z_G < 1$ ); at energies above  $M_P$ , quantum gravity effects would dominate ( $Z_G > 1$ , signaling the saturation of the substrate).

## F. Implications

### 1. Resolution of the Hierarchy Problem

The ratio  $M_P / m_e \approx 10^{22}$  arises purely from geometric attenuation:

$$\frac{M_P}{m_e} = \frac{\alpha^{-\Delta/4}}{\sqrt{B_{res}}} \approx \frac{(137)^{10.75}}{\sqrt{15.3}} \approx 2.4 \times 10^{22} \quad (56)$$

No fine-tuning is required. The hierarchy is exactly the **inverse** of the lattice depth, making gravity’s apparent weakness a consequence of its origin at the bulk centroid.

### 2. The Prohibition of Intermediate Forces

Some theories propose a “Fifth Force” operating at intermediate scales to explain anomalous observations. The  $E_8$ -Persistence Theory prohibits these based on **Topological Stability**. A persistent force requires a stable geometric origin. The lattice excitation possesses only two topologically distinct loci:

- **The Boundary ( $r = 0$ ):** Defines Surface Forces (Gauge fields).
- **The Centroid ( $r = \Delta/2$ ):** Defines Bulk Forces (Gravity).

We conjecture that any signal originating at an intermediate depth (e.g.,  $r = 10$ ) lacks a topological anchor, neither the boundary ( $\chi = 2$ ) nor the centroid ( $\Delta/2$ ), and thus cannot form a stable resonance. Such states would decay into gauge bosons. If this topological stability argument holds, the “Desert” between the Standard Model and Gravity is structurally enforced, prohibiting any stable fifth-force carrier at intermediate energy scales. (*A confirmed detection of a stable intermediate-depth force would falsify this conjecture, as detailed in Section IX*).

### 3. The Planck Scale as a Natural UV Cutoff

The Planck length  $\ell_P = 1/M_P$  represents the resolution floor (Margin) of the geometric substrate. Below this scale, spacetime loses meaning as a continuous manifold, the lattice node addressing becomes discrete. This provides a natural, physical UV cutoff strictly governed by channel capacity, inherently resolving infinities without requiring string theory or loop quantum gravity.

## VII. SYNTHESIS: GLOBAL SOLVENCY OF THE NESTED ARCHITECTURE

In section II, we established that any persistent system whose internal complexity exceeds a single control loop must partition its bandwidth according to three strict

structural rules. Having derived the gauge, scalar, and gravitational sectors of the vacuum independently, we must now demonstrate that they form a mathematically closed, interdependent network. For the  $E_8$ -Persistence theory to be valid, it must be **globally solvent**: the sum of the partitioned subsystems must perfectly exhaust the substrate, the minimal structural remainders must be conserved, and all nested boundaries must seamlessly connect.

#### A. Validation of Rule 1: Capacity Partitioning (The $\nu = 16$ Limit)

Under Rule 1, partitioning is a strict zero-sum allocation of the hardware capacity. The total allocated bandwidth must exactly match the fundamental chiral limit ( $\nu = 16$ ) without overflow or dead space.

By summing the derived capacity loads for the Bulk (Gravity), the Boundary (Topology), and the Gauge (Interaction), we verify the global conservation of information:

$$\underbrace{B_{res}}_{\text{Gravity (Bulk)}} + \underbrace{\frac{\chi}{\sigma - \chi}}_{\text{Matter (Boundary)}} + \underbrace{\alpha}_{\text{Synchronization (Gauge)}} \equiv \nu \quad (57)$$

Substituting the derived geometric values from the previous sections:

$$15.326035981 + 0.666666667 + 0.007297353 = 16 \quad (\text{Exact}) \quad (58)$$

*Note on Circularity:* The sum above is algebraically exact by the definition of  $B_{res}$  as the partition remainder; it cannot fail. The non-trivial claim is cross-sector: the same  $B_{res} \approx 15.326$  independently governs the gravitational depth in Section VI, where it couples with the Beer-Lambert attenuation ( $\alpha^{\Delta/2}$ ) to reproduce both  $\alpha_G$  and  $M_P$  at the correct experimental values. A numerically different remainder — arising from any other partition scheme — would yield incorrect gravitational predictions. The solvency check is therefore a consistency test of the partition structure, not a verification of the sum.

#### B. Validation of Rule 2: Reversible Encoding (The Landauer Floor)

Under Rule 2, a persistent system cannot simply erase geometric mismatches or fractional remainders, as this would incur an infinite dissipative heat penalty (Landauer's Principle). It must strictly conserve this minimal geometric resolution as structural metadata.

The vacuum successfully satisfies this by establishing a strict, non-zero resolution floor: the **Margin Impedance** ( $Z_{MAR}$ ). As derived in Section V E, the geometric dilution of a single unit of information across the lattice's phase-space establishes the electron mass

( $m_e$ ). The architecture does not permit a mass of zero for charged boundaries; the electron represents the exact Landauer Limit of the substrate, the minimum viable topological defect that can be reversibly encoded without dissolving into thermal noise. Rule 1 sets the capacity ceiling ( $\nu = 16$ ); Rule 2 successfully sets the structural floor.

#### C. Validation of Rule 3: Nested Impedance Matching

Under Rule 3, when a system partitions into nested layers, information crossing the boundaries between them must be translated with minimal dissipative loss. This requires the internal processing resistance (Impedance) of the regulatory subsystems to perfectly match the geometric aperture of the signals they govern.

As derived in their respective sections, the vacuum successfully satisfies this requirement at both extremes of the substrate's geometry:

- **The Surface Match (System 5):** The local Higgs regulator structurally matches the aperture of the force it mediates ( $A_{weak} \approx 5.965$ ). Its internal impedance ( $Z_H \approx 5.964$ ) minimizes entropic drag at the  $\chi = 2$  topological boundary, perfectly matching the vacuum resonance to the weak channel capacity.
- **The Core Match (System 6):** The global Gravitational regulator saturates the lattice at the core scale ( $r = \Delta/2$ ). The Planck Mass ( $M_P$ ) is the exact, unique scale required to normalize the bulk geometric attenuation such that the core-to-surface impedance achieves exact unity ( $Z_G \equiv 1$ ).

#### D. Final Verdict: Global Solvency

The Effective Field Architecture of the vacuum is globally solvent. Its  $\nu = 16$  capacity budget is exhausted across the gauge, matter, and gravitational sectors without overflow or dead space. Its structural floor is set by the Landauer limit of the electron, and its nested boundaries are impedance-matched from the topological surface to the geometric core.

The fundamental constants of the Standard Model, the dimensionless couplings (Tier 1), the massive energy scales (Tier 2), and the time-asymmetric causal flow ( $J_{CKM}$ ), are not independent, arbitrary parameters. They are the unique, strictly constrained solutions to a global impedance-matching problem.

The physical universe persists because its substrate transfers information from the lattice core ( $r = \Delta/2$ ) to the surface ( $r = 0$ ) with minimal dissipative loss, reversibly encoding its boundary states (matter) while flawlessly exhausting its finite  $\nu = 16$  capacity budget.

## VIII. EMPIRICAL VALIDATION: THE GLOBAL GEOMETRIC FIT

While Section VII demonstrated that the partitioned architecture is mathematically self-consistent, we must ultimately determine if it describes our physical reality. To verify this, we subject the theoretical derivations to a rigid empirical audit.

### A. Audit 1: The Dynamical Origin of Substrate Friction ( $\eta$ )

To verify the geometric origin of the The Coordinate Overhead ( $\eta \approx 0.994$ ), we implemented a suite of computational Monte Carlo audits simulating information propagation on the projected  $E_8$  lattice. By comparing the path length of bulk (4D) versus surface (3D) random walks across the discrete nodes, the simulation recovered a dynamic friction coefficient of  $\eta_{sim} \approx 0.988$ .

This aligns with the theoretical resonant value ( $\eta_{th} \approx 0.994186$ ) to within 0.7%. This numerical simulation acts as a “Kill-Switch”: it confirms that  $\eta$  is the strict, irreducible geometric cost of projecting a higher-dimensional discrete capacity onto a lower-dimensional coordinate manifold. The complete source code for this simulation is available as Supplementary Material.

### B. Audit 2: The Zero-Degree-of-Freedom Global Fit

As a final empirical validation, we perform a global  $\chi^2$  fit evaluating 9 fundamental observables spanning gauge couplings, electroweak masses, and gravitational scales (including the von Klitzing constant  $R_K$  derived from the geometric impedance in [1]).

Crucially, standard Effective Field Theory global fits achieve their precision by floating continuous input parameters to match the data. The  $E_8$ -Persistence theory possesses **zero floating parameters**. Every theoretical value is rigidly locked to the geometric derivations established in the preceding sections. The sole exception is the QCD running coupling validation in Section IV, which uses the experimentally evaluated vacuum polarization  $\Delta\alpha_{total} \approx 0.0590$  (PDG 2024) as a boundary condition for isolating the geometric overhead  $\eta$ . This empirical input is not a free parameter; it will be derived *ab initio* in the companion paper on fermion mass spectra. All nine observables in the global fit below are purely geometric. The degrees of freedom for this fit is therefore equal to the number of observables ( $N = 9$ ): the statistically conservative choice. A parameterized model with  $p$  free parameters would have  $\chi^2$  distributed with  $N - p$  degrees of freedom, yielding a more favorable reduced  $\chi^2$ . Our zero-parameter constraint therefore *increases* the burden of proof. A reduced  $\chi^2 \gg 1$  would indicate that the

geometric invariants fail to capture the physical correlations;  $\chi^2 \ll 1$  would suggest overfitting or correlated experimental errors. The observed  $\chi^2 \approx 1$  is the unique signature of a structurally determined, non-tunable architecture.

### C. Conclusion of the Audit

A reduced  $\chi^2 \approx 1$  signifies that the theoretical predictions match the experimental observations perfectly within the bounds of expected statistical variance.

Standard physics considers the parameters in Table II to be disjoint properties of reality, requiring individual experimental measurement. The ability of a single geometric lattice to rigidly predict all nine values simultaneously linking the macroscopic gravitational scale ( $M_P$ ), the topological quantization of charge ( $R_K$ ), the breaking of electroweak symmetry ( $G_F, M_W, M_H$ ), and the arrow of time ( $J_{CKM}$ ) into a single unified impedance network, suggests these are not independent fundamental constants. They are the strictly necessary capacity partitions of a single, finite-bandwidth information processing substrate operating in Quiescent Equilibrium.

## IX. STRUCTURAL AUDIT: THE ZERO-DOF CONSTRAINT

The ultimate test of the  $E_8$ -Persistence Theory is replacing the free parameters of the Standard Model global fit. Unlike Effective Field Theories (EFTs) which allow parameters to float to fit data, this framework creates a rigid dependency graph where the inputs are integers.

Consequently, the theory generates specific “Hard Constraints” that serve as falsification criteria. We propose the following analyses for groups with access to electroweak fit packages (e.g., Gfitter, HEPfit).

### A. The Zero-Degree-of-Freedom Electroweak Fit

Standard global fits treat the Fine-Structure Constant ( $\alpha$ ), the Fermi Constant ( $G_F$ ), and the Weak Mixing Angle ( $\sin^2 \theta_W$ ) as floating nuisance parameters constrained only by measurement errors.

**The Audit Task:** Run the global electroweak fit with these parameters fixed to their geometric derivations:

$$\alpha^{-1} = 137.035999212 \dots \quad (\text{Fixed}) \quad (59)$$

$$\sin^2 \theta_W = 0.222905365 \dots \quad (\text{Fixed}) \quad (60)$$

$$G_F = (\sqrt{2}v^2)^{-1} \quad (\text{Fixed via } v_{geo}) \quad (61)$$

**Falsification Criterion:** If the global  $\chi^2$  of the fit degrades by  $\Delta\chi^2 > 9$  ( $3\sigma$ ) compared to the standard floating fit, the geometric invariant hypothesis is falsified. If the  $\chi^2$  remains stable, the free parameters are proven redundant.

## B. Geometric Prohibitions (No-Go Theorems)

The integer invariants  $\{\nu = 16, D = 4\}$  act as hard hardware limits. Unlike effective field theories where new heavy particles can be assumed to exist at higher energy scales, this framework asserts that the **geometry required to host such particles does not exist**.

1. **The Kaluza-Klein Prohibition ( $D = 4$ ):** Standard String Theories require extra spatial dimensions ( $D = 10, 11$ ). The  $E_8$  projection mandates  $D = 4$  to preserve lattice self-duality. *Test:* A confirmed detection of Kaluza-Klein graviton modes or large extra dimensions at the LHC/FCC falsifies the projection mechanism.
2. **The Supersymmetry Prohibition ( $\nu = 16$ ):** The chiral truncation limits the active channel capacity to 16 states per generation. The Standard Model fermions already occupy exactly 48 states ( $3 \times 16$ ). The lattice is saturated. This acts as a double-lock against SUSY. First, there are no available bit-addresses in the node to encode superpartner quantum numbers. Second, extending the manifold with Grassmann coordinates (Superspace) creates an effective fractional dimension that violates the Kneser constraints required for lattice self-duality. The substrate hardware ( $E_8$ ) physically cannot map to the Superspace software. *Test:* The discovery of a supersymmetric partner with quantum numbers occupying the  $\nu = 16$  channel space would require extending the chiral capacity, contradicting the projection derived in Paper 1. Such a discovery would necessitate revisiting the substrate geometry.
3. **The GUT Expectation ( $\sigma \neq \chi$ ):** In this framework, the forces arise from distinct geometric features ( $\Delta$  vs.  $D\Delta$  vs.  $\sigma$ ). Consequently, the theory structurally expects that precision measurements will continue to show the gauge couplings failing to converge at a single energy scale. This is not a strict prohibition, a consistent unified embedding might yet be found, but it identifies the “GUT mismatch” as a natural consequence of the substrate architecture rather than evidence for missing threshold corrections.

## C. The Precision $\alpha$ Vector

As established in [1], the most immediate test of the foundational substrate is the convergence of the Fine-Structure Constant. Because the massive energy scales derived in this nested architecture are geometrically locked to  $\alpha^{-1}$ , deviation in the base impedance physically breaks the global solvency of the hierarchy.

## D. Additional Hard Constraints

**Strong CP and Axion Searches:** The topological prohibition of  $\theta_{QCD} \equiv 0$  (section IV F) predicts that axion searches targeting the Peccei-Quinn mechanism will yield null results and that the neutron electric dipole moment is bounded by the CKM background ( $\sim 10^{-32}$  e-cm). Any confirmed detection of a standard QCD axion or a nEDM measurement exceeding  $10^{-30}$  e-cm would constitute a hard falsification.

**The Prohibition of Intermediate Forces:** The  $E_8$ -Persistence theory allows only two topologically stable force origins: the boundary ( $r = 0$ , gauge forces) and the centroid ( $r = \Delta/2$ , gravity). A stable force originating at an intermediate lattice depth lacks a topological anchor and must decay. This prohibits stable “Fifth Force” carriers at any energy scale between the gauge bosons and  $M_P$  (section VI).

## E. Theoretical Audits (Community Verification)

The framework makes specific structural claims that allow for rigorous theoretical auditing. We identify open derivations that serve as definitive tests:

1. **The Lattice Loop Calculation:** We identified the QCD beta function coefficients (11, 2/3) with geometric invariants of the substrate. An *ab initio* lattice field theory calculation should confirm that these values arise as eigenvalues of the transfer matrix without manual identification.

## X. CONCLUSION: THE CALCULUS OF PERSISTENCE

In this work, we introduced the rules of **Nested Persistence**: Capacity Partitioning, Reversible Encoding, and Impedance Matching. These rules govern any complex system whose internal regulatory demands exceed the capacity of a single control loop. Rather than treating them as qualitative heuristics, we formalized them as a rigorous mathematical calculus for structural stability.

These rules were derived without reference to quantum fields or gauge symmetry. Capacity Partitioning resolves the causal aliasing that would cause an overloaded substrate to decohere. Reversible Encoding ensures that geometric mismatches persist as structural metadata rather than dissipating as heat. Impedance Matching guarantees that information crossing nested boundaries is translated without reflection loss. Each rule is structurally necessary: a system that violates any one of them cannot maintain coherent nested subsystems and will collapse toward a less complex steady state.

To validate this calculus, we applied it to the most fundamental complex system in nature: the quantum

vacuum. By treating the vacuum as a discrete, resource-constrained information-processing substrate (the  $E_8$  lattice derived in previous works), we demonstrated that the fundamental constants of nature are not arbitrary tuning parameters. They are the exact, parameter-free partition coefficients required to allocate a finite bandwidth ( $\nu = 16$ ) across orthogonal geometric interactions.

### A. From Empirical Inputs to Geometric Necessity

The implications for physics represent a fundamental paradigm shift. In standard Effective Field Theory, coupling constants, mixing angles, and mass hierarchies are axiomatic inputs, measured empirically and inserted by hand.

This framework changes that status. We have shown that:

- The **Gauge Couplings** ( $\alpha_s, \sin^2 \theta_W$ ) are strictly the geometric fractions of available channel capacity allocated to spatial structure and temporal updates, and the Jarlskog Invariant ( $J$ ) quantifies the irreducible frustration of projecting 5-fold internal symmetry onto the 4-fold causal manifold.
- The **Mass Scales** (Electroweak and Planck) are the necessary geometric impedance-matching thresholds required to translate signals across nested topological boundaries without dissipative reflection loss.
- The **Strong CP Problem** and the **Hierarchy Problem** are structurally prohibited. They are resolved only by using the strict information-theoretic capacity limits of the substrate hardware.

The success of the zero-degree-of-freedom global fit ( $\chi^2 \approx 1.1687$ ) confirms that the Standard Model is the unique, mathematically exhausted thermodynamic ground state of a finite lattice minimizing Entropic Action.

### B. A Unified Framework for Complex Systems

The physics derivation is ultimately a proof-of-concept for Informational Energetics.

For decades, the study of Complex Adaptive Systems has relied on analogies to explain emergence. We have demonstrated that the foundational theorems of information and control theory—Ashby’s Law of Requisite Variety, Shannon’s Channel Capacity, and Landauer’s Principle—are not merely descriptive analogies; they are the literal, exact constraints that dictate physical reality.

The quantum vacuum represents the **perfect limit of persistence**. Because it is a closed system with no external environment to offset dissipation, it must solve the entropic control problem with zero tolerance for error.

The fundamental constants of nature are the signature of a system operating at maximal informational efficiency. This is why all three rules, all nine measurable constants, and 64 orders of magnitude of the hierarchy fit within three compact papers: the vacuum wastes nothing.

If the vacuum must rigidly partition its bandwidth, reversibly encode its mismatches, and impedance-match its nested boundaries to survive entropic decay, then any system persisting against a less absolute thermodynamic gradient is subject to the same constraints with greater tolerance but identical architecture. IE provides the quantitative blueprint to analyze biological cells, neural architectures, and economic networks as scaled instantiations of this universal architecture of persistence.

### C. Implications for Grand Unification: The Symmetric Precursor

Standard Grand Unified Theories (such as  $SU(5)$ ) predict that at the unification scale, before symmetry breaking, the Weak Mixing Angle converges to  $\sin^2 \theta_W = 3/8 = 0.375$ .

The  $E_8$ -Persistence framework recovers this exact theoretical value naturally, identifying it as the **Unprojected Symmetric State** of the lattice.

Before the lattice undergoes Chiral Truncation to establish the Arrow of Time (which isolates the anti-chiral sector and projects the geometry down to  $D = 4$ ), both  $D_4$  orthogonal sublattices of the  $E_8 \rightarrow D_4 \oplus D_4$  decomposition are fully active, yielding a total manifold rank of  $2D = 8$ . In this symmetric phase, the bandwidth partition is simply the ratio of the Color Sector ( $N_c = \sigma - \chi = 3$ ) to the Total Lattice Rank:

$$\sin^2 \theta_W(M_{GUT}) = \frac{N_c}{2D} = 0.375 \quad (62)$$

The architectural connection is direct: the low-energy Weak Mixing Angle derived previously ( $\sin^2 \theta_W = 43/193 \approx 0.223$ ) is the exact, physical result of projecting this symmetric 3/8 ratio onto the constrained, time-asymmetric 4D causal manifold.

Standard GUTs predict low-energy  $\sin^2 \theta_W \approx 0.21$  at the Z-pole, requiring threshold corrections to reach the observed 0.223. The  $E_8$ -Persistence theory resolves this: the apparent “non-unification” of gauge couplings is the natural consequence of forces arising from distinct geometric features ( $\Delta$  vs.  $D\Delta$  vs.  $\sigma$ ) rather than a single broken symmetry group. The low-energy value is a **fixed geometric ratio** dictated by projection, not a running parameter.

### D. Next Steps: Dynamics and Lifetimes

This paper derived the structural constants governing the vacuum’s nested architecture: the partition coefficients that define which interactions exist and at what

strength. The companion paper on Dynamics applies these fixed partition coefficients to derive how information propagates through the gauge hierarchy—the flavor structure, generational mixing, and the coupling dynamics of matter on the partitioned substrate. The subsequent paper on Lifetimes then addresses how persistent states decay, deriving the fermion mass spectrum as the hierarchy of resonant impedances available to topological defects.

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## Appendix A: The Gauge Admittance: Electroweak Structure and Link Geometry

The derivations in section IV established the dimensionless bandwidth ratios ( $\alpha_s$ ,  $\sin^2 \theta_W$ ), which fix the **relative** allocation of the lattice’s capacity. To complete the physical mapping to the Standard Model, this appendix establishes the **absolute** gauge couplings ( $g$ ,  $g'$ ,  $g_3$ ) that fix the admittance of information flow across the discrete links, and details their physical manifestation as the electroweak gauge bosons.

### 1. The Electroweak Bifurcation

The geometric partition between the temporal resonance ( $\approx 22\%$ ) and the spatial/hardware infrastructure ( $\approx 78\%$ ) physically dictates the entire phenomenology of the electroweak gauge bosons.

The  $E_8$ -Persistence theory establishes a strict structural isomorphism to the Standard Model, where the abstract hypercharge ( $g'$ ) and weak isospin ( $g$ ) gauge couplings are simply the continuous reflections of these discrete lattice bandwidth allocations:

$$\frac{g'^2}{g^2 + g'^2} \longleftrightarrow \frac{\Delta}{N_{sys}} \quad (A1)$$

This geometric split forces the observed behavior of the carriers:

- **The Photon ( $\gamma$ ,  $U(1)$  Temporal):** The hypercharge sector couples strictly to the **Fundamental Resonance** ( $\Delta$ ), representing the causal update rate of the lattice. Because temporal evolution cannot be obstructed without violating causality, the photon carries no spatial hardware load. It must remain massless and propagate at the exact bandwidth limit of the substrate ( $c$ ).
- **The W/Z Bosons ( $W^\pm$ ,  $Z^0$ ,  $SU(2)$  Spatial):** The isospin sector couples to the **Manifold Embedding** ( $N_{sys} - \Delta$ ), representing the dense spatial infrastructure and chiral payload. To propagate through this structural density, these gauge fields must deform the lattice geometry, incurring a severe geometric impedance cost that manifests physically as mass.

### 2. Absolute Couplings from Link Geometry

**The Bare Lattice Coupling:** On the discrete substrate, the link variable  $U_\mu = \exp(-ig\ell A_\mu)$  rotates the phase of transmitted signals. The kinematic bandwidth limit  $c = \ell/\tau = 1$  (Paper 2, Section III.C) enforces exactly one spatial transition per temporal update. The phase rotation per transition is therefore bounded by the geometric step efficiency:

$$g_{bare} = \frac{1}{2} \quad (A2)$$

This is the universal lattice efficiency: half the phase budget per link is allocated to spatial translation, half to internal rotation.

**The Physical Couplings:** The bare admittance is dressed by the specific capacity allocation of each orthogonal sector:

- **Strong coupling  $g_3$ :** The color sector occupies  $N_c = \sigma - \chi = 3$  channels of the  $\nu = 16$  total. The admittance is the geometric mean of the bare coupling and the sector fraction:

$$g_3^2 = 4\pi\alpha_s = 4\pi \cdot \frac{\nu\eta + 1/D}{\alpha^{-1}} \approx 1.54 \quad (g_3 \approx 1.24) \quad (A3)$$

The factor  $4\pi$  arises from solid angle integration in the continuous limit (Paper 2, Section V.B).

- **Weak isospin  $g$ :** The spatial sector couples to the manifold embedding  $N_{sys} - \Delta$ . The admittance is:

$$g^2 = \frac{4\pi\alpha}{\sin^2 \theta_W} \approx 0.42 \quad (g \approx 0.65) \quad (A4)$$

- **Hypercharge  $g'$ :** The temporal sector couples to the fundamental resonance  $\Delta$ . The admittance is:

$$g'^2 = \frac{4\pi\alpha}{1 - \sin^2 \theta_W} \approx 0.12 \quad (g' \approx 0.35) \quad (A5)$$

**Physical Interpretation:** The gauge couplings of standard physics are not independent primitives. They are strictly the **dressed admittances** of the lattice links, renormalized by the capacity allocation of their respective sectors. The “gauge principle” is simply the geometric requirement that phase rotations preserve unitarity across these discrete links.

### 3. Recursive Validation: The Cost of Time

The structural consistency of the theory can be explicitly confirmed by linking this Spacetime Partition directly back to the Fine-Structure Constant.

Recall the **Toll** component of the  $\alpha^{-1}$  geometric impedance ( $T_{geo}$ ), derived from the microscopic 3-point interaction vertex on the substrate:

$$T_{geo} = \frac{1}{(2\nu)^3} \cdot \frac{\chi}{\sigma} \cdot \left(1 - \frac{\sigma}{D\Delta}\right) \approx 1.1852 \cdot 10^{-5} \quad (A6)$$

We now observe that this bare geometric transition cost ( $T_{geo}$ ) structurally mirrors the fully dressed, second-order electromagnetic cross-section ( $\alpha^2$ , as expected for a 3-point vertex) modulated by the Weak Mixing Angle ( $\sin^2 \theta_W$ ) derived above:

$$\begin{aligned} T_{check} &= \alpha^2 \sin^2 \theta_W \\ &= \left(\frac{1}{137.035999212}\right)^2 \cdot \left(\frac{43}{192.906976744}\right) \quad (A7) \\ &\approx (5.325 \times 10^{-5}) \cdot (0.2229053) \\ &\approx 1.18700126 \cdot 10^{-5} \end{aligned}$$

**Conclusion:** The match to within  $\sim 0.15\%$  confirms deep internal consistency. The convergence of these two entirely independent geometric paths—one derived from the local 3-point vertex geometry ( $T_{geo}$ ), the other from the macroscopic bandwidth partitioning ( $T_{check}$ )—demonstrates that the Weak Interaction is strictly the physical manifestation of the temporal bandwidth allocation on the electromagnetic manifold.

## Appendix B: Derivation of Hypercharge (Geometric Flux Conservation)

In the foundational derivations of the  $E_8$ -Persistence substrate [1], the structure of the non-Abelian gauge groups  $SU(3)_C$  and  $SU(2)_L$  emerged strictly from the geometric invariants  $\sigma = 5$  and  $\chi = 2$ . In standard Quantum Field Theory, the corresponding  $U(1)_Y$  hypercharge assignments  $(+1/3, +4/3, -2/3, -1, -2)$  are axiomatic inputs, justified post-hoc by the requirement of anomaly cancellation.

In the  $E_8$ -Persistence theory, anomaly cancellation is **Information Conservation (Unitarity)**. Information (phase flux) cannot be created or destroyed at a lattice

node. By applying this strict geometric flux conservation to the  $\nu = 16$  chiral channels of the substrate, we can derive the exact fractional hypercharge assignments entirely from first principles.

### 1. Step 1: The Channel Partition ( $\nu = 16$ )

The lattice node possesses exactly  $\nu = 16$  chiral channels. These channels are geometrically partitioned by the topological boundary ( $\chi = 2$ ) and the internal bulk ( $\sigma - \chi = 3$ ):

- **Color Partition:** Channels either couple to the internal interaction bulk (Color, multiplicity 3) or they bypass it (Colorless, multiplicity 1).
- **Boundary Partition:** Channels either couple to the active temporal boundary (Left-Handed Doubles, multiplicity 2) or they remain static internal singlets (Right-Handed, multiplicity 1).

This strict geometric intersection partitions the  $\nu = 16$  capacity into exactly the Standard Model fermion architecture:

$$(3 \times 2) + (3 \times 1) + (3 \times 1) + (1 \times 2) + (1 \times 1) + (1 \times 1) = \mathbf{16} \quad (B1)$$

This perfectly generates the quark doublets ( $Q_L$ ), the right-handed quarks ( $u_R, d_R$ ), the lepton doublet ( $L_L$ ), the right-handed electron ( $e_R$ ), and one sterile channel ( $\nu_R$ ), utilizing 100% of the node’s hardware capacity.

### 2. Step 2: The Fractional Denominator (Color Dilution)

Hypercharge ( $U(1)_Y$ ) represents a continuous phase rotation across the node. To preserve the  $SU(3)$  interaction symmetry, any flux entering the color bulk must be distributed isotropically across its dimensions.

Because the color bulk possesses exactly  $\sigma - \chi = 3$  geometric degrees of freedom, the total flux for a color-coupled state must be divided by 3. Therefore, quarks inherently possess fractional hypercharges with a denominator of 3, while color-singlet states (leptons) couple directly to the manifold and carry integer charges.

### 3. Step 3: Geometric Flux Conservation

To preserve Unitarity, the net chiral phase flux (Left-Handed input minus Right-Handed output) must balance to zero across every orthogonal geometric sector.

*a. Boundary Flux Balance:* The total flux traversing the  $\chi = 2$  topological boundary must sum to zero. The three color-coupled doublets ( $Y_Q$ ) must perfectly balance the single colorless doublet ( $Y_L$ ).

$$3Y_Q + 1Y_L = 0 \quad (B2)$$

Because the denominator is strictly fixed to 3 by color dilution, the minimal positive integer unit of flux available to this boundary state is exactly  $+1/3$ . Assigning this fundamental unit ( $Y_Q = +1/3$ ) strictly forces the lepton doublet flux:  $Y_L = -1$ . Geometrically, the Lepton doublet exactly balances the combined flux of the three Quark doublets.

*b. Internal Bulk Balance:* The chiral flux entering the color bulk from the boundary doublets ( $2 \times Y_Q$ ) must equal the flux returning from the right-handed color singlets ( $Y_u + Y_d$ ).

$$Y_u + Y_d = 2Y_Q = \frac{2}{3} \quad (\text{B3})$$

How does the lattice geometrically split this  $2/3$  flux between the two right-handed quark states? Because they are singlets under the boundary, their values must be defined by the remaining orthogonal invariants of the projection: the spatial manifold ( $D = 4$ ) and the boundary reflection ( $-\chi = -2$ ). The constraint  $Y_u + Y_d = 2/3$  combined with the requirement that numerators be geometric invariants of the projection ( $\{1, \chi, D, \sigma\}$ ) admits a unique solution:  $Y_u = D/3 = 4/3$  and  $Y_d = -\chi/3 = -2/3$  perfectly satisfies this conservation law:

$$Y_u + Y_d = \frac{D - \chi}{3} = \frac{4 - 2}{3} = \frac{2}{3} \quad (\text{B4})$$

*c. Total Manifold Balance:* Finally, the net chiral flux across all active channels on the macroscopic manifold must be zero ( $\sum_{LH} Y - \sum_{RH} Y = 0$ ). We established that the Left-Handed boundary sum is zero. Therefore, the Right-Handed output sum must also be zero:

$$3Y_u + 3Y_d + 1Y_e = 0 \quad (\text{B5})$$

Substituting the derived quark fluxes ( $3(4/3) + 3(-2/3) = 4 - 2 = 2$ ), we solve for the right-handed electron:

$$2 + Y_e = 0 \implies Y_e = -2 \quad (\text{B6})$$

Geometrically,  $-2$  is exactly  $-\chi$ . The right-handed electron carries the exact inverse flux of the topological boundary.

#### 4. Conclusion: The Derived Charge Structure

This geometric framework naturally recovers the hypercharge assignments without requiring supplementary anomaly cancellation conditions. By strictly enforcing information conservation (Unitarity) across the  $\nu = 16$  chiral channels, the lattice geometry mathematically forces the exact fractional charges of the Standard Model.

Table III summarizes this derivation, demonstrating that every fermion's hypercharge is identically the geometric ratio required to conserve flux between the manifold ( $D = 4$ ), the color bulk ( $\sigma - \chi = 3$ ), and the topological boundary ( $\chi = 2$ ).

#### 5. Derivation of Higgs Hypercharge

As established in System 5, the scalar sector acts as the regulatory bridge anchored to the lattice Unitary Ground State ( $\Delta^0 = 1$ ). Geometrically, this mandates a unitary hypercharge for the scalar doublet:  $Y_H = +1$ .

This purely geometric assignment perfectly recovers the observed physical charge states of the Higgs field. Evaluating the Gell-Mann–Nishijima relation ( $Q = I_3 + Y/2$ ) with this geometric hypercharge ( $Y_H = +1$ ) and the  $SU(2)_L$  isospin components ( $I_3 = \pm 1/2$ ) yields exactly the neutral ( $H^0$ ,  $Q = 0$ ) and charged ( $H^\pm$ ,  $Q = \pm 1$ ) components of the Standard Model Higgs doublet.

#### Appendix C: Data Availability

The geometric invariants  $\{11, 2/3\}$  emerge as eigenvalues of the discrete transfer matrix on the  $\mathbb{S} = \{D = 4, \sigma = 5, \chi = 2\}$  tangent space. Computational audits verifying this emergence without manual identification are available at <https://github.com/meta-r0ze/Informational-Energetics-Nested-Persistence>.

Table II. The Zero-DOF Global Geometric Audit. The combined fit across both tiers demonstrates a reduced  $\chi^2 \approx 1.1687$ . Because there are zero tunable parameters, this precision strictly indicates that the geometric constraints of the  $E_8$  lattice natively encode the physical correlations of the Standard Model. Tier 3 values are not included in the global  $\chi^2$  calculations. Because they are strict algebraic consequences of the Tier 1 and Tier 2 inputs, including them would double-count degrees of freedom. They are presented here to demonstrate the framework's dependent predictive power.

| Observable                                       | Geometric Value                                  | Experimental Value                                              | Deviation             |
|--------------------------------------------------|--------------------------------------------------|-----------------------------------------------------------------|-----------------------|
| <i>Tier 1: Bosonic Structure &amp; Couplings</i> |                                                  |                                                                 |                       |
| $\alpha^{-1}$                                    | 137.035999212                                    | $137.035\,999\,177 \pm 0.000\,000\,021$ [14]                    | $1.68\sigma$          |
| $G_F$                                            | $1.166\,378\,82 \times 10^{-5} \text{ GeV}^{-2}$ | $(1.1663788 \pm 0.0000006) \cdot 10^{-5} \text{ GeV}^{-2}$ [10] | $0.04\sigma$          |
| $M_W$                                            | $80.384\,578\,879 \text{ GeV}$                   | $80.377 \pm 0.012 \text{ GeV}$ [10]                             | $0.63\sigma$          |
| $\alpha_s(M_Z)$                                  | 0.117903156                                      | $0.1179 \pm 0.0009$ [9]                                         | $0.00\sigma$          |
| $M_H$                                            | $125.269\,263\,77 \text{ GeV}$                   | $125.20 \pm 0.11 \text{ GeV}$ [10]                              | $0.63\sigma$          |
| <b>Tier 1 <math>\chi^2</math></b>                |                                                  |                                                                 | <b>3.2366 (5 DOF)</b> |
| <i>Tier 2: Mass Hierarchy &amp; Resistance</i>   |                                                  |                                                                 |                       |
| $\sin^2 \theta_W$                                | 0.222905365                                      | $0.222\,91 \pm 0.000\,11$ [10]                                  | $-0.04\sigma$         |
| $J_{CKM}$                                        | $3.08489953 \cdot 10^{-5}$                       | $(3.08 \pm 0.15) \cdot 10^{-5}$ [10]                            | $0.03\sigma$          |
| $M_P$                                            | $1.220\,893\,55 \times 10^{19} \text{ GeV}$      | $(1.22091 \pm 0.00001) \cdot 10^{19} \text{ GeV}$ [14]          | $-1.65\sigma$         |
| $R_K$                                            | $25\,812.807\,469\,41 \Omega$                    | $25\,812.807\,45 \pm 0.000\,01 \Omega$ [14]                     | $1.94\sigma$          |
| <b>Tier 2 <math>\chi^2</math></b>                |                                                  |                                                                 | <b>7.2815 (4 DOF)</b> |
| <b>Combined Total <math>\chi^2</math></b>        |                                                  |                                                                 | <b>10.5181(9 DOF)</b> |
| <b>Combined Reduced <math>\chi^2</math></b>      |                                                  |                                                                 | <b>1.1687</b>         |
| <i>Tier 3: Derived Consequences</i>              |                                                  |                                                                 |                       |
| $\lambda$                                        | 0.12942366                                       | $0.129 \pm 0.005$ [10]                                          | $0.08\sigma$          |
| $y_e$                                            | $2.93422684 \cdot 10^{-6}$                       | $(2.935 \pm 0.001) \cdot 10^{-6}$ [10]                          | $-0.77\sigma$         |
| $\alpha_G$                                       | $1.75179958 \cdot 10^{-45}$                      | $(1.752 \pm 0.001) \cdot 10^{-45}$ [14]                         | $-0.20\sigma$         |
| $\beta_0$                                        | $11 - \frac{2}{3}n_f$                            | 11 (exact)                                                      | Exact                 |
| $\rho_{vac}/M_P^4$                               | $1.18103720 \cdot 10^{-123}$                     | $(1.38 \pm 0.20) \cdot 10^{-123}$ [10]                          | Consistent            |
| $v$                                              | $246.219\,637\,823 \text{ GeV}$                  | $246.219\,65 \pm 0.000\,06 \text{ GeV}$ [10]                    | $-0.20\sigma$         |

Table III. Derived Fermion Hypercharges via Geometric Flux Conservation

| Particle             | Lattice Partition  | Multiplicity | Derived $Y$ | Geometric Ratio         |
|----------------------|--------------------|--------------|-------------|-------------------------|
| Quark ( $Q_L$ )      | Boundary & Color   | 3            | $+1/3$      | $+1/(\sigma - \chi)$    |
| Lepton ( $L_L$ )     | Boundary           | 1            | $-1$        | $-3 \times Y_Q$         |
| Up Quark ( $u_R$ )   | Manifold & Color   | 3            | $+4/3$      | $+D/(\sigma - \chi)$    |
| Down Quark ( $d_R$ ) | Reflection & Color | 3            | $-2/3$      | $-\chi/(\sigma - \chi)$ |
| Electron ( $e_R$ )   | Reflection         | 1            | $-2$        | $-\chi$                 |